

Prague University of Economics and Business
Faculty of Informatics and Statistics



The Impact of Basketball Statistics on Winning a Game: Comparison of Czech NBL, Euroleague and NBA

BACHELOR THESIS

Study program: Matematické metody v ekonomii

Specialization: Ekonometrie a operační výzkum

Author: Štěpán Srp

Supervisor: Mgr. Vladimír Holý, Ph.D.

Prague, May 2023

Acknowledgements

I would like to thank my supervisor, Mgr. Vladimír Holý, Ph.D., for his consultations, insights, and advices while writing this thesis. I would also like to thank all of my family, for their help with this thesis and their support throughout my studies.

Abstract

This thesis uses logistic regression to analyse which basic basketball statistics are important in determining the outcome of a game. The thesis compares six seasons of the Czech National Basketball League (NBL), Euroleague, and the NBA, both in the regular season and playoff, using only basic box score statistics or their derivatives. The created models were based on three different sets of predictors: the first set contained only offensive/defensive efficiency rating, the next set consisted of the so-called Four Factors, and the last one fitted ordinary statistics obtainable directly from box scores. The results suggest that efficient shooting, turnovers, and rebounding are among the most important factors to winning games in all leagues, with each league differing in which factors are more important. In the NBA, there was a larger emphasis on efficient shooting, the 3-point shot was more important than in other leagues, with an apparent upward trend over time. The Czech NBL and Euroleague tend to be more strategically oriented, emphasizing turnovers and rebounds more than in the NBA. Additionally, the European leagues showed less difference between playoff and regular season, while in the NBA, the difference in play style between the two was larger. In the NBA playoff, the game becomes more similar to Euroleague and Czech basketball, with a greater emphasis on rebounding, steals, and turnovers.

Keywords

basketball, logistic regression, basketball statistics, comparasion

Contents

Introduction	9
1 Logistic Regression	10
1.1 Model	10
1.2 Maximum Likelihood Method	11
1.3 MLE in Logistic Regression	12
1.4 Estimates Interpretation	12
1.4.1 Standardized Coefficients	12
1.4.2 Statistical Significance	13
1.5 Assumptions	13
2 Leagues' Specifics	14
2.1 Czech National Basketball League	14
2.2 Euroleague	15
2.3 National Basketball Association	15
2.4 Comparison of Rules	16
3 Hypothesis and Expectations	17
4 Data	18
4.1 Data Collection	18
4.2 Data Cleaning	19
4.3 Data Description	19
5 Models	21
6 Results for Regular Season	23
6.1 Model 1 - Offensive and Defensive Rating	23
6.1.1 Residual Analysis	23
6.1.2 Results	24
6.2 Model 2 - Four Factors	25
6.2.1 Residual Analysis	26
6.2.2 Results	27
6.3 Model 3 - Basic Statistics	28
6.3.1 Multicollinearity Analysis	28
6.3.2 Results - Assists	29
6.3.3 Results - Rebounds	29
6.3.4 Results - Steals and Turnovers	31
6.3.5 Results - Field Goal Efficiency	31
6.3.6 Home Advantage	32
6.4 Trends in Coefficients	34

6.4.1	Coefficient for 3PT%	34
6.4.2	Turnovers and steals coefficients	35
6.4.3	Offensive rating	36
7	Results for Playoff	38
7.1	Model 1 - Offensive and Defensive rating	38
7.1.1	Results	39
7.2	Model 2 - Four Factors	40
7.2.1	Results	41
7.3	Model 3 - Basic Statistics	41
7.3.1	Results - Residual Analysis	42
7.3.2	Results - Field Goals	42
7.3.3	Results - Rebounds	42
7.3.4	Results - FT%	43
7.3.5	Results - Turnovers	43
8	Discussion	45
	Conclusion	46
	Bibliography	48
A	Appendix A	53
A.1	Transformation of coefficients for Model 3 in Regular season	53
A.2	Table of Results for Logistic Regression Analysis of Model 1, Regular Season and Playoff	54
A.3	Table of Results for Logistic Regression Analysis of Model 2, Regular Season and Playoff	55
A.4	Table of Results for Logistic Regression Analysis of Model 3, Regular Season and Playoff	56
A.5	Proof for variable selection	57

List of Figures

1.1	Logistic function	11
6.1	Analysis of Accuracy from all models fitted for Regular Season	24
6.2	Residuals from Czech and NBA models, based on Four Factors predictors	26
6.3	3PT% Coefficient development between season 2016/2017 and 2021/2022 for NBA	35
6.4	Turnover and steal coefficient development between season 2016/2017 and 2021/2022 in Euroleague and NBA	36
6.5	<i>ORtg</i> coefficient development between season 2016/2017 and 2021/2022 in Euroleague and NBA	37
7.1	Analysis of Accuracy from all models fitted for Playoff	39
7.2	Logistic Regression Analysis of Four Factors on Winning a Game in Regular Season and Playoff	40
7.3	Logistic Regression Analysis of Basic Statistics on Winning a Game, Regular Season and Playoff	44

List of Tables

6.1	Analysis of games with high residuals for ORtg and DRtg models	25
6.2	Model 1 - ORtg	25
6.3	Model 2 - Logistic Regression Analysis of Four Factors on Winning a Game in Regular Season	27
6.4	Model 3 - Logistic Regression Analysis of Basic Statistics on Winning a Game in Regular Season	30
6.5	Share of 2-point shots attempted between 2016/2017 and 2021/2022 seasons . . .	32
6.6	Winning percentage for home games in the three leagues	34
7.1	Model 1 - Logistic Regression Analysis of Offensive/Defensive rating on Winning a Game in Playoff	38
7.2	Average number of possessions in Regular Season and Playoff	43
7.3	Mean score difference in Playoff	43
A.1	Model 3 transformed - Logistic Regression Analysis of Basic Statistics on Winning a Game in Regular Season	53
A.2	Logistic Regression Analysis of ORtg and DRtg on Winning a Game - comparison of Regular Season and Playoff	54
A.3	Model 2 - Logistic Regression Analysis of Four Factors on Winning a Game, Regular Season and Playoff	55
A.4	Logistic Regression Analysis of Basic Statistics on Winning a Game - comparison of Regular Season and Playoff	56

List of abbreviations

2PT%	2-point success percentage	FTR	Free-throw rate
3PT%	3-point success percentage	NBA	National Basketball Association
AS	Assist	OLS	Ordinary Least Square
BL	Block	OR	Offensive rebound
CI	Confidence intervals	OR	Odds Ratio
Czech NBL	Czech National Basketball League	OREB%	Offensive rebound percentage
DR	Defensive rebound	ORtg	Offensive rating
DRtg	Defensive rating	ST	Steal
eFG%	Efficient Field Goal Percentage	TO	Turnover
EURO	Euroleague	TPP	Turnovers per possession
FT%	Free-throw success percentage	VIF	Variance Inflation Factor

Introduction

Coach, players, sport analytics, general managers are all trying to find a recipe for success in basketball competitions. The game itself is influenced by many various factors, the most apparent ones are being recorded as statistics into „box scores“. It is simple for anyone, with basic basketball knowledge, to recognize which of these statistics are good or bad for a team. What is not as simple, is to calculate or quantify how much each statistic influences the outcome of a game. It is impossible for a human brain alone to recognize how much a single rebound, assist or foul influence the team's odds of winning a game.

The game of basketball itself is a complex set of actions and reactions to these actions. It is therefore difficult or almost impossible to create a full model, which gasps every aspect of the game. This thesis aims to simplify the complex model into only few actions captured by basic statistic and tries to approximate the effect of each one.

It uses logistic regression to compare the strength of different factors (e.g. statistics), to determine which are more or less important to winning a basketball game. Its purpose is not to create a prediction model able to predict the outcome of a game, rather than to identify, after the game ends, the key statistics (actions) which helped the given team win. For more context, it provides comparison of regular season and playoff in 3 different leagues – Czech National Basketball League, Euroleague and the NBA. They each represent a different level and style of play. It establishes few hypotheses based on literature and author's personal experience.

There are many basketball factors that influence the outcome of a game. In the top leagues, there are huge amounts of data collected each game. However, the less prestige the league is, the less amount of data are collected. However, in most of the leagues a basic standard set of statistics is always collected. These statistics allow us to examine and quantify, what happened during that game. It also allows us to compare the different leagues with different rules, skill levels and technologies, because these basic statistics are unified across the world.

The aim of this thesis is to describe factors influencing the outcome of a game using only the basic box score of a game. It tries to analyse the differences in factors across the leagues and using various literature and author's personal experience as a player, coach, and a basketball fan, it tries to put the results of the analysis into context of each league, with regard to different play style, rules, and competition system in each league. Apart from academical purposes, this thesis can work as material for players and coaches as a way to adapt to different styles in different leagues and prepare better for each league's specifics.

1. Logistic Regression

A logistic regression model is a model that estimates the probability of an event occurring. It uses a dichotomous response variable, where the proportions of each group are treated as probability of an event occurring. In this case it is not suitable to use linear regression models based on OLS method, as there are 3 main problems with OLS (DeMaris, 1995):

- the predicted values from OLS are outside a range 0 and 1,
- when a binary response variable is used in OLS, the exogeneity assumption is violated,
- the error term is inherently heteroskedastic as variance is $\pi(1 - \pi)$ (DeMaris, 1995).

Therefore, a logistic regression model is used instead of an OLS method and Maximum Likelihood Method is used for estimation of parameters.

1.1 Model

Instead of a linear function, a standard logistic function is used. The standard logistic function is sigmoid curved, approximately linear in the middle, but curved at both ends, as shown in Figure 1.1. The basic equation to describe the outcome of a logistic regression uses the standard logistic function:

$$\hat{Y} = \frac{e^u}{1 + e^u},$$

where $\hat{Y}$ is the estimated probability, and u is the usual linear regression equation:

$$u = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_k X_k = \boldsymbol{\beta} \mathbf{X}^T$$

with constant β_0 , coefficients β_j and predictors X_j for k predictors ($j = 1, \dots, k$). And $\mathbf{X}$ is a matrix of predictors and $\boldsymbol{\beta}$ is a vector of coefficients.

This formula ensures, that the right-hand side is always bound between 0 and 1. This model is linearized, using a well-behaving properties of natural logarithm (DeMaris, 1995). The estimated probability is converted to the log odds, therefore the right-hand side needs to be converted as well. The result is as follows:

$$\ln \left(\frac{\hat{Y}}{1 - \hat{Y}} \right) = \boldsymbol{\beta} \mathbf{X}^T,$$

where $\ln \left(\frac{\hat{Y}}{1 - \hat{Y}} \right)$ is the log odds of an event happening. The model is now analogous to the linear regression model. Now the model can be estimated via Maximum Likelihood Method.

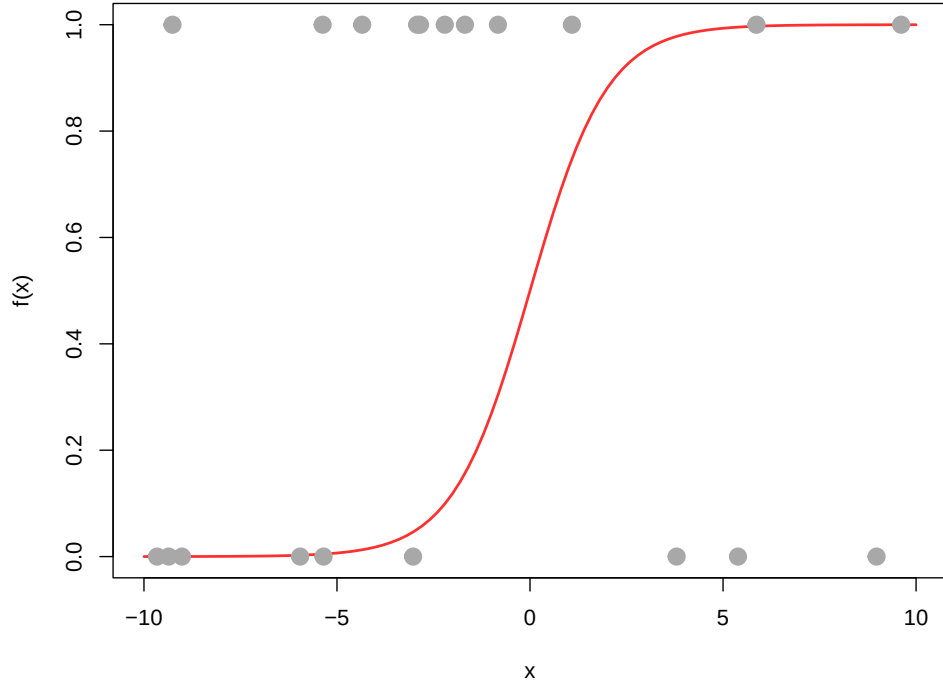


Figure 1.1: Logistic function

1.2 Maximum Likelihood Method

Maximum likelihood method (MLE) is one of the general approaches to estimation. If $Y_1, Y_2, \dots, Y_n$ denotes a random sample from the population distribution $f(y; \theta)$, then the likelihood function is defined as (Wooldridge, 2013):

$$L(\theta; Y_1, \dots, Y_n) = f(Y_1; \theta) f(Y_2; \theta) \cdots f(Y_n; \theta).$$

This is a random variable because it depends on the outcome of a random sample $Y_1, Y_2, \dots, Y_n$. The maximum likelihood estimator of θ , is the value of θ that maximise the likelihood function. The maximum likelihood principle says that, out of all the possible values for θ , the value that makes the likelihood of the observed data largest should be chosen (Wooldridge, 2013):

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \prod_{i=1}^n f(Y_i; \theta),$$

where Θ is a parameter space.

Usually, it is more convenient to work with a log-likelihood function. Since the logarithmic function is monotonous, the position of maximum in the MLE does not change. We use the fact that the log of the product is the sum of the logs:

$$\ln[L(\theta; Y_1, \dots, Y_n)] = \sum_{i=1}^n \ln[f(Y_i; \theta)].$$

Maximum likelihood estimation is under certain regularity conditions consistent, asymptotically normal and asymptotically efficient (Wooldridge, 2013).

1.3 MLE in Logistic Regression

Since we model probability in logistic regression, the estimated probability in the model has Bernoulli distribution, so we model the likelihood function as:

$$L(\beta) = \prod_{i=1}^n P(Y_i = y_i) = \prod_{i=1}^n \frac{[\exp(-x_i' \beta)]^{1-y_i}}{1 + \exp(-x_i' \beta)}. \quad (1.1)$$

However, the parameters of the logistic regression can not be estimated algebraically and need to be estimated numerically using iteration method such as Newton Raphson Method. The estimated values of the logistic regression relate to the log-odds and not to the expected value, as it is with linear regression (Zlámel, 2013).

1.4 Estimates Interpretation

The estimated coefficients ($\hat{\beta}$) from the logistic regression are so-called log-odds, their interpretation is problematic, but they provide an easy way to recognize, which predictors influence the outcome variable positively (log odds having positive values) and which negatively (log odds having negative values).

In order to better interpret the coefficients, a transformation to odds ratios is used. This is done by $OR = e^{\beta_i}$, where β_i is coefficient for the i -th variable. The odds ratios are then defined as odds of the outcome occurring ($\hat{Y}$) versus not occurring ($1 - \hat{Y}$) for each independent variable. The interpretation is that 1-unit change in the independent variable multiplies the odds of the outcome by the amount of e^{β_i} (Stoltzfus, 2011-10)

1.4.1 Standardized Coefficients

When dealing with predictors with vastly different scales or when the scale would be difficult to interpret, it is proposed by (Wooldridge, 2013) to use *standardized coefficients*. This way, the standardized coefficients measure the *standard deviation* change of odds (log-odds, respectively). The formula for standardization is:

$$Z = \frac{X - \mu}{\sigma},$$

where X is the raw score, μ is the population mean and σ is the population standard deviation.

1.4.2 Statistical Significance

The statistical significance is tested using the *Wald test*, where the Wald-statistic is computed as (Tabachnick and Fidell, 2013):

$$W_i = \frac{\beta_i^2}{SE_{\beta_i}^2} \sim \chi_1^2$$

where $SE_{\beta_i}^2$ is the squared standard error of β_i .

1.5 Assumptions

Generally, logistic regression has fewer assumptions than the linear regression. It has 4 assumptions which need to be fulfilled before the analysis.

Linearity in the logit Logistic regression assumes a linear relationship between continuous predictors and the logit transform of the dependent variable. There are many methods to test this assumption, with one of the simplest being the Box-Tidwell test. In this approach, interaction terms between the predictor and its natural logarithm are added to the regression. If any of those interaction terms are significant, the assumption is violated (Tabachnick and Fidell, 2013).

Absence of Multicollinearity Logistic regression is sensitive to high correlation between the predictors. High correlation is often signalled by large standard errors. In this thesis, the multicollinearity is tested using Variance Inflation Factor (VIF) statistic (Tabachnick and Fidell, 2013).

Absence of Outliers Outliers, similar to linear regression, can cause poor fit and poor prediction from the logistic regression. Outliers are often found using residual analysis, where large residuals signalize an outlier. Apart from visual residual analysis, Cook's distance is also used to find out any outliers in the model (Tabachnick and Fidell, 2013).

Independence of Errors The logistic regression assumes that all outcomes are separate of each other. If one's data were to include repeated measures or other correlated outcomes, errors will be similarly correlated, and the assumption would be violated. This assumption is tested visually from the residuals (Tabachnick and Fidell, 2013).

2. Leagues' Specifics

This thesis aims to compare the influence of different statistic on an outcome of a game. It compares the effects of regular season and the playoff. It also compares different leagues. For the comparison, 3 leagues have been chosen and analysed. The leagues chosen were:

- Czech National Basketball League (Czech NBL)
- Turkish Airlines EuroLeague (Euroleague)
- National Basketball Federation (NBA)

The motivation behind choosing these 3 leagues was to compare the style of play in different parts of the world and on different skill levels. The NBA is widely regarded as the best basketball league in the world, it also represents the „American“ style of basketball. The Euroleague is considered the second-best league in the world and the best league in Europe. It presents the best of „European“ style of basketball. While the two aforementioned leagues are continent wide, the Czech league serves as a represent of a national league. Competition held only on territory of one state, in this case, Czech Republic. It is considered to be on a substantially lower level than the NBA and Euroleague. Based on their prestige and geographical factors, each league has its specifics, which need to be described in order to comprehend the results of this thesis in wider sense and understand its findings in the context of each league.

2.1 Czech National Basketball League

Czech National Basketball League is the highest basketball league in the Czech Republic. It is only a national league, meaning it is only played inside the territory of the Czech Republic and only Czech teams are allowed to participate. However, a portion of the league consists of foreign players, mainly from the USA and former Yugoslavia countries (Minarovičová, 2014).

There are 12 participating teams, each playing a regular season consisting of 11 games; each team playing twice with every other team in the league, once home, once away. From there, the first 8 teams qualify to „Winners stage“, where each team plays an additional 14 games. The other 4 teams qualify to „Losers stage“, where each team plays 12 additional games. The last two teams from the „Winners stage“ and the first 2 teams from the „Losers stage“ compete in a play-in games for two playoff spots (BASKETBALL, 2021).

Once playoff seeding is set, quarter-final playoff begin, with the best seeded team playing the worst seeded team, second best playing the second worst and so on. Quarter and semi-final playoff rounds are played till one team has 4 wins (BASKETBALL, 2021).

The best 2 teams advance to the finals, which are played till 3 wins of one team. Also, a 3rd-place playoff round is played for the losers of semifinals series, these are played until one

team has 2 wins (BASKETBALL, 2021).

2.2 Euroleague

Euroleague is a Europe-based competition aiming to put together the best teams of Europe to compete against each other. It serves as an additional competition to the national competitions, where the teams are competing as well. Euroleague is considered the second-best basketball league in the world and the best basketball league in Europe. That is because only the best teams on the continent are able to qualify and allowed to play in the Euroleague.

In the 2022/2023 season there were 18 teams competing in the Euroleague. The regular season is played with the same system as the Czech NBL, where all 18 teams play each other twice for a total of 34 regular season games. There is no „Winner“ or „Looser stage“ in the Euroleague and the first 8 teams after the regular season automatically advance to the playoff (*Competition Format 2022-23*, 2022).

In the first round of the playoff, 8 teams play best-of-five series. The matchups are determined the same way as in Czech NBL. The winner of the series advances to the Final Four. There, each team plays only one semifinal game and only one final game (or 3rd place game for the loser of the semifinals) (*Competition Format 2022-23*, 2022).

2.3 National Basketball Association

The NBA is widely considered the best basketball league in the world. It is viewed as a national competition, but its scale is larger than any other supra-national league. Unlike Czech NBL and Euroleague, the NBA is a closed competition, meaning that there is a fixed number of teams and no matter the final standings, no team can be demoted to a lower league. This is however possible in the Czech NBL and Euroleague.

There are 30 teams in the NBA, each playing 82 regular season games. The teams are divided into two conferences, which are then divided in divisions. The regular season format is as follows:

- 4 games against the other 4 division opponents,
- 4 games against 6 (out-of-division) conference opponents,
- 3 games against the remaining 4 conference teams,
- 2 games against teams in the opposing conference. (*How the NBA Schedule is Made*, 2017-05-08)

After the regular season ends, the first 8 teams from each conference qualify to the playoff¹. The seeding matchups are the same as in both aforementioned leagues. There, 3 rounds

¹Since 2020/2021 season, a play-in tournament to determine the 7th and 8th seed has been used after the

of best-of-seven series are played to determine the winner of each conference. The winners of each conference meet in the NBA Finals, to play in a best-of-seven series (*NBA playoffs*, 2023-04-17).

2.4 Comparison of Rules

Each league is specific with the number of teams participating, number of total games and the different playoff formats. The leagues also differ in basketball rules. There are some rule differences between the NBA and the European basketball. While in Europe, the rules are unified under FIBA organization, the NBA has a different set of rules not unified with FIBA (*Differences between NBA and FIBA rules*, 2022-08-05). The basic and elementary rules are the same in all three leagues, however there are some important differences, between the NBA and Europe. These deviances in rules, even though small and for a casual fan not important, can help understand the differences in factors determining the outcome of a game in the results of this thesis.

Court size The NBA court is slightly wider and longer.

Three point range While in the NBA the three point line is 7.24 meters away (The distance is shorter in the corner) in Europe, the distance is everywhere the same: 6.75 m.

Game duration In the NBA every quarter is 12 minutes long, in the FIBA competitions, the length of one quarter is 10 minutes.

Goaltending and interference There are different rules for interference with a basket after a shot attempt. Generally, in the NBA the rules about touching the basketball after a shot is in the air are stricter, than in Europe. This rule is effecting mostly rebounds and blocks.

Defensive 3 second rule While in both competitions staying under the basket, in the restricted area, on offence for more than 3 second is prohibited, in the NBA the same rules applies to defence as well. This limits the defence and opens up opportunities to score more easily (*Differences between NBA and FIBA rules*, 2022-08-05).

regular season and before the first round of the playoff.

3. Hypothesis and Expectations

The analytic part of the thesis focused on analysis of basketball statistics. This analysis was carried out to identify and describe effects various statistics have on winning/loosing basketball games. Before the analysis itself, a various hypothesis and/or expectations were established. These hypotheses were based on literature, previous studies, sport-analytical articles and author's personal experience.

The hypotheses and expectations were:

- **Impact of „good“ and „bad“ statistics:** At the minimum, it is expected that all the basic statistics generally perceived as „good“¹ will have positive impact on winning and statistics perceived as „bad“² will have negative impact on winning games.
- **Increase in defensive importance in playoff:** A various NBA players have said that the intensity in playoff increases (Redick and Alter, 2023-03-30). Combined with the first point, the expectation is that the importance of defensive statistics will increase in playoff in all leagues (especially in the NBA).
- **The rising importance of 3 point shot over time in the NBA:** The game of basketball has changed over time and more shots are now coming from beyond the three-point line (Schuhmann, 2021-10-14). This mainly applies to NBA, which seems to be the pushing this trend forward. It is therefore expected, that the three point shot will be increasingly more important.
- **Greater emphasis on 3-point shot in the NBA:** Similar to the previous point, it is expected that 3-point shot will be of a greater value in the NBA than in the other two leagues.
- **Assists more important in Europe than in NBA:** NBA and European basketball are played differently. Based on author's experience and (Mandić et al., 2019-10-07) it seems that European basketball is more tactically oriented and teams rely on teamwork rather than on individual performances, like in the NBA. It is expected that assists will play a larger role in winning a game in Europe-based competitions than in the NBA.

¹These are: 3PT%, 2PT%, FT%, DR, OR, AS, ST, BL, F-OP

²These are: TO and F-minus

4. Data

For the purpose of this thesis, data from 3 basketball leagues were downloaded and analysed. In this chapter, the data will be described more closely, so that the result in the upcoming chapters can be comprehended in the context of the original data.

4.1 Data Collection

This study analysed data from 3 basketball leagues: National Basketball Federation (NBA), Euroleague and Czech National Basketball League (Czech NBL). Data were analysed from the 2016-2017 through 2021-2022 season (6 seasons). Earlier seasons were not included due to change in competition format in Czech NBL starting from 2016-2017 season. Also, longer time period could be contra-productive, as basketball style of play everywhere evolves and factors determining the outcome of a game could change in time.

Each league has different formats of both regular season and playoff, and different number of teams, resulting in varying number of games played each season. Therefore, the most observations overall were gathered from NBA, then Euroleague and the least amount of observation from Czech NBL.

For the NBA, consisting of 30 teams, each playing 82 games per regular season, the sample size for the given time period was 7059 games. For Euroleague consisting of 15–18 teams ¹, each playing 30 or 34 games each season, sample size was 1498 games. For Czech NBL where 12 teams are participating, each playing 22 games per regular season, the sample size was 789 games.²

For playoff, the number of observations was varying based on different playoff formats in each league. Again with most observations collected from NBA ($n = 498$) followed by Euroleague ($n = 114$) and Czech NBL ($n = 138$)

A test and train datasets were created from the original data to test the accuracy of the models. Data were split randomly into the two datasets, with train set containing 70% of the original data and test set containing 30% of the original data. These sets were then used to compute confusion matrices for each model to verify their predictive capabilities. The results presented are from models trained on the full datasets (including both train and test datasets). This was done for each league and for both regular season and playoff.

Data were collected from Czech Basketball Federation (*CZ.BASKETBALL*, 2023), Hack A Stat website, which is aggregating data from official Euroleague box scores (*Advanced stats*

¹Till 2018-2019 there were 16 teams, from 2019-2020 until 2020-2021 - 18 teams, 2021-2022 - 15 teams

²Note that during 2019-2020 season COVID-19 pandemic had its impact on all three leagues, resulting in shorter regular seasons and no playoff

for everybody, 2023) and for NBA a package in statistical software R „nbastatR“ (Bresler, 2023) was used.

4.2 Data Cleaning

Data were downloaded and processed using statistical software R Studio (R Core Team, 2021). Data were cleaned and appropriately transformed. Since each season was downloaded separately, all seasons were bind together for each league. In the downloaded data, there were duplicates present, as every game was recorded twice; once from the perspective of a home team and once from the perspective of an away team. These records were bound together, so that each game is recorded as one row, with 1 team's statistics labelled as „Team A“ statistics and other team's statistics labelled as „Opponent's“ statistics. Same games were paired via unique „Game ID“, which was part of the original NBA data and was created for Euroleague and Czech NBL during the data cleaning.

Also, other statistics not included in the downloaded data sets were added. These statistics were computed from the original statistics and are further described in the following chapter.

4.3 Data Description

The original data sets contained many statistics, but only a few were used in the final models. Some variables were slightly transformed, but are still considered basic statistics, as they are clearly extracted from the datasets itself. Basic statistics used in models were:

- **Win** - binary variable, indicating whether team won ("1") or not ("0")
- **3PT%** - 3-point success percentage, calculated as $3PT\% = 3PT_{Made}/3PT_{Attempted}$
- **2PT%** - 2-point success percentage, calculated as $2PT\% = 2PT_{Made}/2PT_{Attempted}$
- **FT%** - Free-throw success percentage, calculated as $FT\% = FT_{Made}/FT_{Attempted}$
- **DR** - Defensive rebound
- **OR** - Offensive rebound
- **AS** - Assist
- **TO** - Turnover
- **ST** - Steal
- **BL** - Block
- **Foul** - Foul of a team, i.e. how many times did the team foul the opponent

Additional statistics were computed and added to the data as well. These statistics were not part of the original data, but were computed afterwards and used in the analysis. Most of them are based on concept of possessions. Possessions are defined by (Kubatko et al., 2007-01-09) as:

„A possession starts when one team gains control (or possession) of the basketball and ends

when that team gives up control of the basketball. Teams can give up possession of the basketball in several ways, including (1) made field goals or free throws that lead to the other team taking the ball out of bounds, (2) defensive rebounds, and (3) turnovers. Note that under this definition of a possession, an offensive rebound does not start a new possession; an offensive rebound starts a new play.“

However, possessions are not part of a classical/basic box score and can not be calculated from the classical box score. Therefore, they needed to be estimated. For estimation of possessions for each team in each game, a formula by (Kubatko et al., 2007-01-09) was used to estimate possessions

$$Poss_t = FGA_t + 0.44 \cdot FTA_t - OR_t + TO_t,$$

where FGA is field goals attempted, FTA is free-throws attempted. The constant 0.44 is empirically found constant, estimating the proportion of possession-ending free-throws³ in a game as found by (Kubatko et al., 2007-01-09).

Based on possessions, more advanced statistic were added to the datasets. These advanced statistics were computed based on (Kubatko et al., 2007-01-09). The statistic were:

- **ORtg** - team’s efficiency on offence. How many points does a team score compared to the number of possessions they had. Calculated as: $ORtg = PTS_t / Poss_t \cdot 100$.
- **DRtg** - team’s efficiency on defence. How many points does a team allow the opponent to score compared to the number of possessions the opponent had. Calculated as: $DRtg = PTS_o / Poss_o \cdot 100$.
- **eFG%** - Field goal success percentage, that accounts for 3 pointers made. Calculated as: $eFG\% = (FG_{Made} + 0.5 \cdot 3PT_{Made}) / FGA$.
- **TPP** - Turnovers per possession, calculated as: $TPP = TO / Poss$.
- **OREB%** - Offensive rebounding percentage. How many offensive rebounds team collects out of all offensive rebounds available in a game. Calculated as: $OREB\%_t = OR_t / (OR_t + DR_o)$.
- **FTR** - Free-throw rate. How large proportion of all shots in a game consist of successful free-throws. It demonstrates the team’s ability to get fouled and earn free-throws. Calculated as: $FTR = FTM / FGA$.

The last four statistics (eFG%, OREB%, TPP, FTR) create so-called *Four Factors*; a set of statistics used in basketball analysis to provide team’s summaries of overall performance on per-possession basis. These factors are calculated for both offence and defence, the difference between a given offensive and defensive factor, indicates whether a team „won“ a factor (Kubatko et al., 2007-01-09).

³Possession ending free-throws are all free-throws except first free-throws of two, first and second free-throws of three, or free throws due to technicals, flagrant fouls, and clear path fouls. Also free-throws after made field goals are not (double) counted, since the made field goal already has counted that possession. In the 2002-03 through 2005-06 seasons, 43.8% of free throws were possession ending free throws (Kubatko et al., 2007-01-09).

5. Models

Data were analysed for any missing or error values. A multiple logistic regression models were then fitted to the data. The dependent variable was *Win*. Independent variables varied across models, as different models tried different approaches and different sets of variables to explain the dependent variable. It is worth noting that the purpose of all models is not predicting the outcome of games, as it is common with logistic regression. The aim of this thesis is to identify factors contributing to winning (or loosing) a basketball game. Variables included are often known only after the given game ends and are alone subject to various predictions. Therefore, if any of these models were to be used to predict the outcome of a game before the game actually begins or even during the game, these models would have performed poorly, as they are not meant for prediction.

Models for each league, both regular season and playoff, were created. Overall, 18 models were fitted. One section of this thesis is always dedicated to comparison of 3 of those models fitted on the same set of predictors. For intelligibility of the 3 sets of variables used, these sets are referred to as:

Model 1:

$$\pi[Win] = \frac{1}{1 + \exp(-\beta_1(ORtg_t - ORtg_o))},$$

where index t is team t and index o is their opponent.

Model 2:

$$\pi[Win] = \frac{1}{1 + \exp(-b_1)},$$

where:

$$b_1 = \beta_1(eFG\%_t - eFG\%_o) + \beta_2(TPP_t - TPP_o) + \beta_3(OREB\%_t - OREB\%_o) + \beta_4(FTR_t - FTR_o),$$

where index t is team t and index o is their opponent.

Model 3:

$$\pi[Win] = \frac{1}{1 + \exp(-b_2)},$$

where:

$$b_2 = \beta_1(3PT\%_{Diff}) + \beta_2(2PT\%_{Diff}) + \beta_3(FT\%_{Diff}) + \beta_4(ST_{Diff}) + \beta_5(DR_{Diff}) + \beta_6(OR_{Diff}) + \beta_7(AS_{Diff}) + \beta_8(TO_{Diff}) + \beta_9(Foul_{Diff}) + \beta_{10}(BL_{Diff}),$$

where index *Diff* means difference in the given statistic between the team t and their opponent's o . Therefore, $3PT\%_{Diff} = 3PT\%_t - 3PT\%_o$ etc.

The difference between the team's statistic and their opponent statistic, was computed to satisfy $P[Win_t] = 1 - P[Win_o]$. If this modification was not done, it can be proved that the former equation would not stand. More information can be found in Appendix A.5.

Also, the constant was removed from all models, as the probability of a team winning when all statistics are 0 is 0. By removing the constant, the model is forced to start at probability of winning 0, for all predictors equal to 0. All models include a statistic indicating scoring of a team. If this statistic is 0, it means the team did not score and therefore can not win any game.

For readability purposes, the predictors in the results of this thesis (such as tables and figures) are named after the general name/abbreviation of the static, rather than by the full interaction term. E.g. the variable inputted in the regression ($3PT\%_t - 3PT\%_o$), is in the results labelled simply as $3PT\%$.

For models 1 and 2, the data were standardized using z-scores, so that final regression coefficients were standardized (Tabachnick and Fidell, 2013). In model 3, original data were used for easier interpretation of results. For playoff, the same 3 models were constructed (respectively, the same 3 sets of predictors were used).

Also, a regression models based on data from only single season were created for all 3 sets of predictors and all observed seasons. The aim was to check for any trend in regression coefficients in time and try to interpret the coefficient change in time. This is discussed in Chapter 6.4.

6. Results for Regular Season

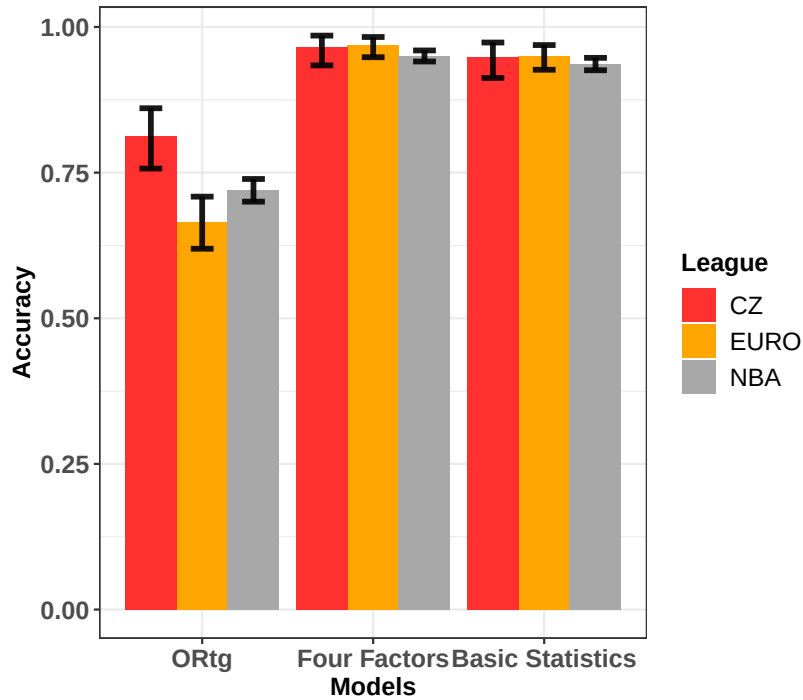
6.1 Model 1 - Offensive and Defensive Rating

The first model examines the effect of offence (defence, respectively) overall on winning games. Offence and defence are wide terms and can mean a number of different activities on the court. For this model, offence and defence have been reduced to Offensive rating (efficiency), respectively Defensive rating (efficiency) of a team, described in Chapter 4.3. As described in the previous chapter, these ratings are measures of team efficiency, i.e. how many points does a team score in regard to the number of possessions they had in a game. Only the offensive rating has been included in the model, as from the nature of this statistic and the given dataset, the defensive rating coefficient would have had the same absolute value. The variable was standardized using z-scores. A linearity in the logit was tested through Box-Tidwell approach (Tabachnick and Fidell, 2013). No statistically significant relationship between the predictor and the logit of the outcome variable has been discovered ($p > 0.1$). Since only one independent variable entered the model, there was no need to check for multicollinearity. Analysis of residuals confirmed independence of errors for all three leagues, this was to be expected from the nature of the original data. Even though some remote observations occurred, none exceeded Cook's distance of 1. Accuracy from the confusion matrix was not high, as it was below 0.82 for all three cases (Figure 6.1). This problem is discussed in the upcoming section.

6.1.1 Residual Analysis

Residuals in all models were independent, but in each model there was at least one outlier present. None of the outliers were influential, but the models showed a pattern of not predicting specific types of games correctly. The games with the highest residuals are showed in Table 6.1.

There is a pattern for models based on *ORtg* (resp. *DRtg*) of not predicting close games well. All three games, which had the largest residuals, had a difference in score not larger than 2 points (Table 6.1). Also, in two of the three games, the winning team was trailing for the first three quarters and came back in the fourth quarter. Also, in all three cases the losing teams had better *ORtg* than the losing team and had won in almost all basic statistics. In Euroleague and NBA, the only statistics which were in favour of the winner were offensive rebounding and turnovers. In Czech league the winner were better in offensive rebounding, 3-point percentage and assists. It seems that *OR* was the determining factor in close games where this model(s) failed to predict the outcome.



Note: Error bars = 95% CI

Figure 6.1: Analysis of Accuracy from all models fitted for Regular Season

6.1.2 Results

Results of the 3 regression models are shown in Table 6.2. Using *model chi-square* test, all three models were statistically significant with $\chi^2 < 0.01$ (DeMaris, 1995). Also, the variable *ORTg* was statistically significant. From the nature of the variable *ORTg* and *DRTg* and their use in regression, both offensive and defensive efficiency are equally important. Comparing the standardized coefficients, the Europe-based competitions seemed to value efficiency more. In Euroleague the emphasis on making the most out of the team's possessions is greater than in the other two leagues, especially compared to the NBA. There, the emphasis on scoring on most of the team's possessions is the weakest, this might be due to different style of play in the NBA. This is further discussed later in the text.

Also, as mentioned earlier, the prediction capability, expressed by the accuracy in Figure 6.1, is below 0.82 in all cases. Even though the variable *ORTg* carries the information about the final score of the game, none of the models, based solely on this predictor, could predict the outcome of a game well enough. This suggests that winning basketball games is more about just scoring efficiently, and there are more factors involved in winning a basketball game.

Table 6.1: Analysis of games with high residuals for ORtg and DRtg models

League	Winning Team	Losing Team	Diff. in score	Statistics by winner	won	Note
Euroleague	Bamberg	Valencia	+1	more OR, less TO		Winner trailing until 3Q
NBA	Rockets	Warriors	+1	more OR, more ST, less TO		Winner trailing until 3Q
Czech NBL	Prostějov	Ústí nad Labem	+2	more OR, better 3PT%, more AS		-

Table 6.2: Model 1 - ORtg

<i>Dependent variable: Win</i>			
	Czech NBL (95% CI)	Euroleague (95% CI)	NBA (95% CI)
ORtg	9.098*** (6.889, 11.307)	10.781*** (8.883, 12.679)	8.282*** (7.695, 8.870)
Observations	789	1,498	7,059
Log Likelihood	-58.981	-105.204	-698.895
Akaike Inf. Crit.	119.962	212.407	1,399.790

Note:

*p<0.1; **p<0.05; ***p<0.01

6.2 Model 2 - Four Factors

The second model was explaining variable *Win* with co called *Four Factors*, both offensive Four Factors (statistics of a given team) and defensive Four Factors (statistics for the opponent). Offensive and defensive factors were subtracted from each other in an interaction. Four Factors are explained in Chapter 4.3. All independent variables were standardized using z-scores. Box-Tidwell transformation showed no violation of linearity of logit assumption (Tabachnick and Fidell, 2013). No other assumptions were violated, with $eFG\%$ and $eFG\%_{OP}$ having borderline value of VIF however, always below 8.

6.2.1 Residual Analysis

The analysis of residuals showed some outliers, mainly in NBA and Czech NBL, these are shown in Figure 6.2. None of the outliers were influential, and none exceeded Cook's distance of 1. Individual games with high residuals were analysed, and some patterns were observed among them:

- All games were close in score, with the highest difference in score being 5 points.
- Almost all games were played in the beginning of the season, most of them before the New Year.
- The losing teams were better in the majority of basic statistics than the winning team.
- All winning teams had significantly more offensive rebounds (even though they had less defensive rebounds).
- In the NBA and Czech NBL apart from offensive rebounds, turnovers were also the statistics in which the winners were better.
- In Euroleague every game with high residuals had common participant - *CSKA Moscow*.
- In Czech NBL the same was observed with *Sluneta Ústí nad Labem*.

There were various characteristics that were connecting all the games with high residuals. With offensive rebounds and turnovers being the most apparent connectors. Also, a pattern of common team showing across all games with high residuals was found in Czech NBL and Euroleague. This might be due to specific play style of the teams, environment in which the games were played or other factors, such as rest the teams had between the games.

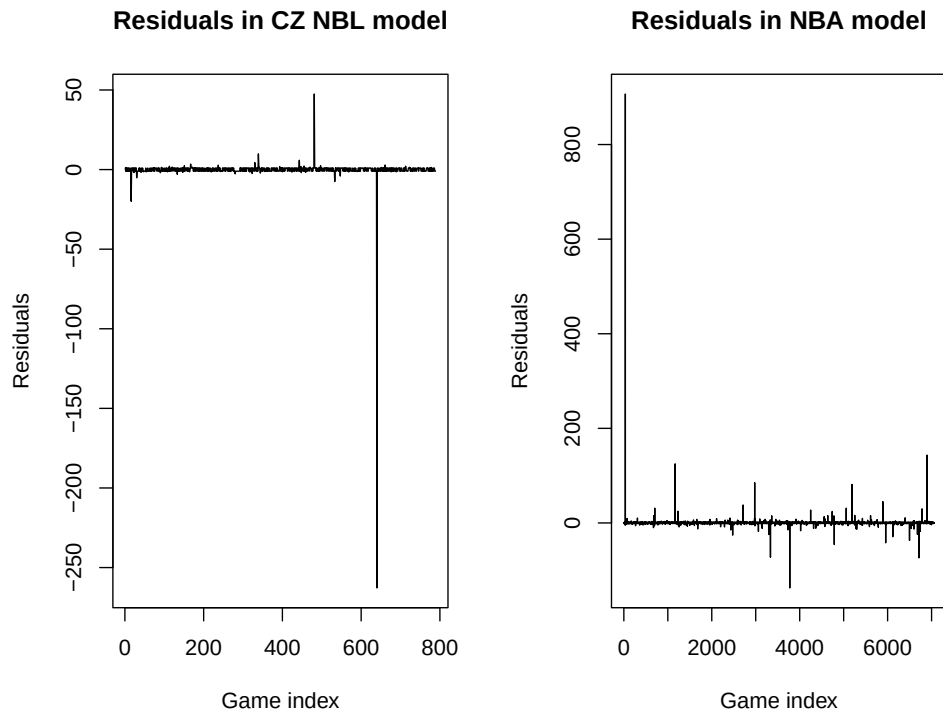


Figure 6.2: Residuals from Czech and NBA models, based on Four Factors predictors

6.2.2 Results

The most significant factor for winning games in all three leagues was efficient shooting; having high efficiency of offence or forcing opponent to an inefficient shooting percentage on defence. (Table 6.3).

Table 6.3: Model 2 - Logistic Regression Analysis of Four Factors on Winning a Game in Regular Season

	<i>Dependent variable: Win</i>		
	Czech NBL $\hat{\beta}$ (95% CI)	Euroleague $\hat{\beta}$ (95% CI)	NBA $\hat{\beta}$ (95% CI)
eFG%	7.347*** (5.499, 9.195)	6.068*** (5.181, 6.955)	6.819*** (6.347, 7.291)
TPP	-4.304*** (-5.444, -3.164)	-3.734*** (-4.334, -3.134)	-3.275*** (-3.520, -3.031)
OREB%	3.405*** (2.459, 4.350)	3.056*** (2.560, 3.551)	2.749*** (2.540, 2.958)
FTR	1.506*** (1.006, 2.005)	1.623*** (1.306, 1.940)	1.775*** (1.625, 1.925)
Observations	789	1,498	7,059
Log Likelihood	-57.754	-157.818	-749.386
Akaike Inf. Crit.	123.508	323.635	1,506.771

Note:

*p<0.1; **p<0.05; ***p<0.01

The most important factor was efficient shooting (*eFG%*), which was by far the most important to winning games. This could be expected, as without scoring points, no team can win a game. The second-highest coefficient was *TPP*. It seems that the ability to take care of the basketball and not commit turnovers is of a great value to winning games. However, the amount of effect it had differed across the three leagues. It was by far the lowest in the NBA, where committing turnovers was not costing the team chance for victory nearly as much as it did in Europe or Czech Republic. This can be due to different style of play and game pace in NBA and Europe overall. While in Europe, basketball is more tactical, individual, and possessions are longer, in NBA the focus is more on fast-paced athletic basketball with less focus on tactical offence, resulting in shorter possessions (Mandić et al., 2019-10-07). This can also be backed up by the data collected. While on average, an NBA team had slightly

more than 2 possessions per minute played (more than 2.1 Poss/Minute), a team playing in Euroleague or Czech NBL had an average of less than 2 possessions per minute played (less than 1.95 Poss/Minute in Czech NBL, less than 1.8 Poss/Minute in Euroleague).

Similarly, predictor *OREB* was not as influential, in the NBA, to winning a game as in Europe-based leagues (Table 6.3). Again, this could be explained by the style of play, as (Mandić et al., 2019-10-07) suggested, that it is a tactical point not to commit players to offensive rebounds in the NBA. On the contrary, a higher free-throw rate increased the odds of winning a game more in the NBA compared to Euroleague, this could be explained by different style of referring a game in different leagues. Another possible factor suggested by (Mandić et al., 2019-10-07) is that players in Euroleague are better prepared to reduce the number of free-throws per foul. Therefore, committing „smarter“ fouls without awarding opposing team with free-throws. This could be true to Czech NBL as, in author’s personal experience, players in youth categories in the Czech Republic are being taught to commit „smarter“ fouls, not awarding opponent’s with free-throws.

Compared to the first model, which included only the *ORtg* variable, in these models the accuracy of prediction was significantly higher (Accuracy > 0.94 in all cases). This is confirming, that more predictors than just offensive (defensive) efficiency are important in determining the outcome of a basketball game.

6.3 Model 3 - Basic Statistics

As Table 6.4 shows, the third model included all basic statistics as independent variables. In this case, no variables were standardized, this was made in order to better interpret the results of the logistic regression. Log Odds are presented in Table 6.4, more easily interpretable odds ratios can be found in Appendix (Table A.1). All models were tested for linearity in logit, multicollinearity, absence of outliers and independence of errors. Linearity in logit assumption was not violated, also no outliers were located in the solution and errors seemed to be independent of each other.

6.3.1 Multicollinearity Analysis

A moderate positive correlation between variable *AS* and variables *2PT%*, *3PT%* was detected during analysis in all three models. This correlation apparently roots in the general definition of an assist, as with each assist a made 2-point or 3-point shot is associated. However, the correlation between the two variables was not considered very strong (for both measured correlations $\rho < 0.45$ across all three models). Also, a moderate negative correlation was detected in NBA model between *2PT%* and *OR*. This, again, could be explained by the nature of the offensive rebound, which can only be achieved by missing a shot attempt. The more shots are missed (meaning lower shot success percentage), the more offensive rebounds are available (and collected). Interesting is the fact that moderate correlation is only

between 2-point shot percentage and offensive rebounds, and not between 3-point shot percentage and *OR*. Suggesting that in the NBA, unsuccessful 2-point shots are more likely to be rebounded by the offence. Nevertheless, since all mentioned correlations are only moderate and subsequent VIF analysis showed no worrisome results, all the discussed variables were left in all three regression models.

6.3.2 Results - Assists

The results of LR analysis of basic basketball statistics on winning a game are presented in Table 6.4. All models were significant for predicting a game outcome ($\chi^2 < 0.01$). Also, all predictors except blocks, assists, and steals were statistically significant. Non significance of assists' beta's is unexpected. As various researches suggested, assists are important to team's long-term and short-term success (Lorenzo et al., 2019; Ibáñez et al., 2008-11). One would then expect assists to be an important factor in winning a basketball game. The reason for assists not being influential to winning a game in model 3 is the aforementioned correlation between *AS* and *2PT%*, *3PT%*. This is causing unexpected value of coefficients for *AS*, which contradicts the literature and common belief. In the NBA the assist coefficient is statistically significant, however, its effect is the smallest out of the all statistically significant variables.

6.3.3 Results - Rebounds

Similar to the previous model in Chapter 6.2, rebounds play an important role in winning games. Both offensive and defensive rebounds are among the most important predictors in all three leagues. However, offensive rebounds seem to play a larger role in determining the outcome of a game. In the Europe-based competitions, offensive rebounds are the predictors, which increase the odds of a team winning the most. In the NBA, the importance is notably lower and overall shooting is similarly or even more important (Table 6.4).

The same can not be said about the defensive rebounding. Firstly, defensive rebounds were overall less important. Both 2-point shot success and 3-point shot success were more important in all three leagues. Secondly, the coefficients were more similar (usually the estimates of other two leagues are within the 95% CI of the given estimate). Also, the lowest importance had a defensive rebound in Czech NBL, not the NBA. What stayed the same, was that in Euroleague, the importance of securing defensive rebounds was the highest of all three leagues.

Overall, it seems that in Europe, securing rebounds is more important than in the NBA. This might go hand in hand with a different style of play in America. In the NBA, players are more physically gifted, i.e. NBA players are on average 4 cm taller than Euroleague players (Webb et al., 2012). Meaning, it is easier for them to grab a rebound, therefore they are not as worthy or important to winning a game. Combined with the already mentioned supposed tactical decision not to commit players to offensive rebounding in the NBA, defensive rebounds are

Table 6.4: Model 3 - Logistic Regression Analysis of Basic Statistics on Winning a Game in Regular Season

	<i>Dependent variable: Win</i>		
	Czech NBL $\hat{\beta}$ (95% CI)	Euroleague $\hat{\beta}$ (95% CI)	NBA $\hat{\beta}$ (95% CI)
OR	0.492*** (0.342, 0.643)	0.472*** (0.377, 0.567)	0.376*** (0.340, 0.412)
DR	0.231*** (0.093, 0.368)	0.322*** (0.213, 0.431)	0.287*** (0.251, 0.323)
BL	-0.010 (-0.164, 0.145)	0.062 (-0.044, 0.168)	0.005 (-0.030, 0.039)
AS	0.057 (-0.024, 0.138)	0.035 (-0.018, 0.089)	0.057*** (0.036, 0.079)
ST	0.022 (-0.121, 0.165)	0.086 (-0.018, 0.189)	0.116*** (0.072, 0.160)
TO	-0.746*** (-0.954, -0.539)	-0.798*** (-0.944, -0.653)	-0.561*** (-0.612, -0.510)
Fouls	-0.267*** (-0.370, -0.165)	-0.234*** (-0.300, -0.168)	-0.276*** (-0.303, -0.248)
2PT%	0.398*** (0.289, 0.506)	0.328*** (0.268, 0.389)	0.388*** (0.356, 0.419)
3PT%	0.362*** (0.271, 0.453)	0.330*** (0.278, 0.381)	0.374*** (0.348, 0.399)
FT%	0.077*** (0.045, 0.109)	0.065*** (0.046, 0.085)	0.078*** (0.068, 0.087)
Observations	789	1,498	7,059
Log Likelihood	-83.665	-205.576	-1,031.040
Akaike Inf. Crit.	187.330	431.153	2,082.080

Note:

*p<0.1; **p<0.05; ***p<0.01

not as worthy there as they are in Europe.

6.3.4 Results - Steals and Turnovers

Even more important than rebounds were turnovers. Turnovers measure the ability to keep the ball during possession and end the possession with a shot, rather than losing the ball without attacking the opponents' basket. Again, circling back to the style of play in each league, the most costly were turnovers in Europe. There, each turnover multiplied the odds of winning a given game by approximately 0.45. On the other hand, in the NBA turnover multiplied the odds of winning by 0.57.

As discussed in Chapter 6.2.2, the NBA has a different style of play, focusing on high tempo basketball, even at the cost of lots of turnovers. In Czech NBL and Euroleague, the focus is more on tactical basketball, executing each play correctly and valuing every possession (Mandić et al., 2019-10-07). Therefore, every turnover is hurting the team's odds of winning more in Europe-based competitions.

What is surprising is the fact that, except for the NBA, steals were not statistically significant predictors. One would expect that if turnovers are hurting a team's odds of winning a game, steals would have a similar, but positive, effect. Here, it is worth noting that every steal is a turnover for the other team, but not every turnover is a steal, as a steal is counted only if the team on defence regains control of the basketball without stopping the game (e.g. by having to inbound after an out-of-bound play). Therefore, while turnover automatically means success for the opponent's defence in such way that opponent lost control of the basketball without endangering the basket. The importance of a steal itself would then be somewhat hidden in the turnover statistic of the opponent. The added importance of a steal, compared to turnover, is that the ball is „alive“ and there is potential for a fast-break scoring opportunity.

6.3.5 Results - Field Goal Efficiency

Field goal efficiency was the key to winning a basketball game in the previous model. Here, the rebounds seemed to be more important, however 2-point shot, 3-point shot and free-throw efficiency all positively impacted team's odds of winning a game. The least influential was free-throw percentage, which was in all three observed leagues similar and lowest of the three efficiency predictors (Table 6.4). This might be due to the fact that free-throws are worth only one point and that they are overall easier to make, so teams in all leagues have almost mastered them and therefore the percentage is always high and similar. Two point shot percentage coefficients were ever so slightly higher than for the 3-point shot in the NBA and Czech NBL, and almost identical in the Euroleague. That is because 2-point shots were still on average more common shots in all three leagues (Table 6.5).

Interestingly, the two point shot was the most important to winning a game in Czech NBL, where the share of the 2-point shot is the lowest among the three leagues. In Czech NBL, with 1 percentage point improvement in 2-point percentage, the odds of winning a game were multiplied by 1.48. In Euroleague the importance of the 2-point shot was the lowest. This might be, as discussed before, due to team play and other statistics than only shooting percentage playing an important role.

In the NBA, the 3PT% coefficient was the largest. This was to be expected as various literature described, the main shot location in the NBA is shifting from 2-point shots toward 3-point shots (Freitas, 2021-06-01), meaning the 3-point success percentage is becoming more and more important. The NBA, with its fast-paced style, is considered the leader in this change. It can be expected that the margin will gradually decrease overtime in the other two leagues as well. The trend of 3-point shot percentage importance is further described in the upcoming chapter.

Table 6.5: Share of 2-point shots attempted between 2016/2017 and 2021/2022 seasons

League	Share of 2pt-shots attempted
Czech NBL	0.604
Euroleague	0.613
NBA	0.641

Note: Free-throws are not included

6.3.6 Home Advantage

At first, the importance of home court wanted to be observed. However, including a binary variable indicating whether a team played at home or not, led to violation of multiple assumptions of the logistic regression. However, it has been shown that playing at home is important. Even though it is not directly included in the logistic regression models, this section of the thesis will be devoted to *Home court advantage*. A simple analysis has been conducted to determine how many games does a team win at home.

A home court advantage is when a team wins more than 50% of their games at home. It has been shown, that home court advantage, in the NBA, is among the strongest of all major team sports in the USA (Entine and Small, 2008-04-28). There are 4 main sources of the home court advantage (Nevill and Holder, 1999-11-01):

- crowd factors,
- familiarity with local conditions,
- travel factors,
- effects related to rule difference.

An effect regarding the referees subconsciously favouring the home team has also been described in various papers, however the opinions differ and there are no conclusive results

(Dohmen and Sauermann, 2016-09-01; Nevill and Holder, 1999-11-01).

Playing at home was the most important in the Euroleague, where home team has won almost 62% of their games, followed by Czech NBL with 61% home game win percentage and NBA with 56.6%. Differences between the leagues might be due to factors, mentioned above. This thesis does not empirically conclude which factors are more or less responsible for the home court advantage in each league, but given the specifics of each league, it tries to come up with possible explanations regarding the different home win percentages at home (Tab. 6.6).

In Euroleague, the teams have to travel not only for its national competition but for Euroleague matches as well. The travel distance is multiple times larger than inside one country, with the average team travelling more than 40 000 km each season (*EuroLeague in distance*, 2022). Taking the distance and the time zone change into account, it is not surprising that home court increased the odds of winning the biggest of all three leagues. Playing at home can mean that players are more rested and ready to perform at their highest level and contribute to winning a game. It was actually shown that both length of the trip and number of days between games are not dramatic but notable and statistically significant factor (Entine and Small, 2008-04-28).

In the Czech NBL, the benefit of playing at home was not as big. That was due to the fact that an average Czech NBL team travelled around 3600 km per regular season. That is more than 10 times less than an average Euroleague team. Czech players do not have to deal with long travels, varying time zones and fatigue connected with both of those. Secondly, the average attendance was significantly lower in Czech NBL compared to Euroleague or NBA. This means that both travel and crowd effects were reduced in Czech NBL environment, therefore the strength of home court advantage was reduced.

In the NBA, the effect was the weakest of the three. Even though an average NBA team travels more than 64 000 km per season; most of the three (McMahan, 2018-10-26) and also has the biggest game attendance (Gough, 2022-06-23). The NBA, however, is an extreme case, where teams, in the same time span, play more than double the games than the teams in other leagues. This mean that in the NBA, a team has on average 1.13 day of rest between games and almost double the distance to travel compared to European teams. This can very well cancel out any strong home court advantage, as there is just simply not enough time to recover and rest even when teams play at home. Of course, there is a positive home court advantage effect, as the NBA schedule is optimized to limit travelling and prolong rest between games (Gröer, 2017-02-15). Also with the massive attendance to NBA games ¹, it is likely that crowd in NBA are behind home court advantage in NBA more than in other leagues. On the other hand, travel factors including rest factors might be behind the home advantage in Euroleague.

¹17 000 fans are on average attending an NBA game vs 8800 fans are on average attending an Euroleague game (Gough, 2022-06-23; *EuroLeague registers all-time attendance, competitiveness records at mid-season / EuroLeague*, 2023-01-18)

Table 6.6: Winning percentage for home games in the three leagues

League	Win percentage
Czech NBL	0.6096
Euroleague	0.618
NBA	0.566

6.4 Trends in Coefficients

An analysis of coefficients' evolution in time for each model was carried out to identify any trends in coefficients. This was done using logistic regression on the same data as previous analysis. However, the data were split into individual seasons and a separate model was estimated for each of the seasons. For each league, there were 6 different models (1 for each season) for each set of predictors². Coefficients' trends were analysed visually and the most interesting findings are presented in this section. However, only coefficients for Euroleague and NBA were analysed in the end. For Czech NBL, there were not enough observations and in the models perfect separation occurred, causing the coefficients to be abnormally inflated.

6.4.1 Coefficient for 3PT%

The most significant trend was found with variable 3PT% from NBA dataset. Compared to the other leagues, where no apparent trend was observed, the NBA showed significant positive trend in coefficient of 3PT% (Figure 6.3). This is supported by various literature, which states that NBA teams are gradually putting more emphasis on 3-point shots (Freitas, 2021-06-01). If this linear trend is sustained in the upcoming seasons, the 3-point shot success percentage will soon be more important to winning games in the NBA than the 2-point shot.

In Czech NBL, the coefficients were inflated in certain season (season 2017/2018, 2021/2022) as well as had large variance, where the 3PT% coefficient was vastly different each season with no signs of a trend or fluctuation around a given number. This might be explained by the flow of foreign players in the Czech highest league. It can be assumed that the better the league, the harder it is for a player to sign a contract with a team in this league. The NBA and Euroleague are considered the best two basketball leagues in the world, therefore a smaller less prestige Czech NBL is easier to get to and the flow of the players coming and leaving the Czech league would be higher. As (Minarovičová, 2014) noted, basketball in the Czech Republic is a secondary sport and Czech NBL is attracting players from countries such as the USA or former Yugoslavia, where basketball is considered a primary sport. These foreign players are then responsible for the majority of points scored (BASKETBALL, 2022-10-28; BASKETBALL, 0209-08-10). This mean that foreign players are „dictating“ the type of offence played in Czech league. So, with different foreign players fluctuating in and out

²There were 3 sets of predictors/variables based on predictors from the three models mentioned in Chapter

of Czech league, the dominant offensive style in the league changes, explaining why 3-point percentage importance varies from season to season.

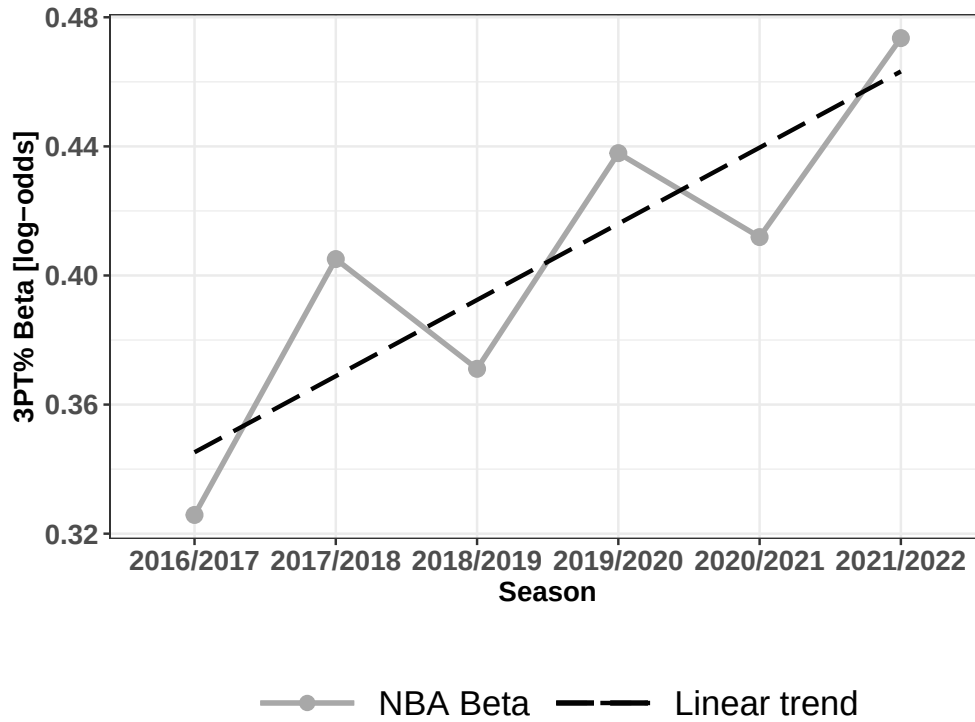


Figure 6.3: 3PT% Coefficient development between season 2016/2017 and 2021/2022 for NBA

6.4.2 Turnovers and steals coefficients

Turnovers and steals parameters in time were also analysed for Euroleague and NBA. Both of the coefficients for Euroleague showed a trend of losing importance in recent years (Figure 6.4). Even though steals' coefficient in Euroleague, was non-significant in some models based on only one season, its development was still included. While in the NBA, both coefficients showed little variance and seemed to fluctuate around a given value, the Euroleague coefficients varied across the season without a clear upward or downwards trend. This might be due to smaller number of observations in each Euroleague season. Another explanation might be that the NBA is more closed competition and if players change teams it is usually inside the NBA. On the other side, Euroleague is more open, both with players, who fluctuate in and out of the competition more often and teams, as Euroleague teams can be promoted to or demoted from Euroleague. Therefore, depending on the combination of players and teams currently in the Euroleague, the importance of the statistics might differ from season to season.

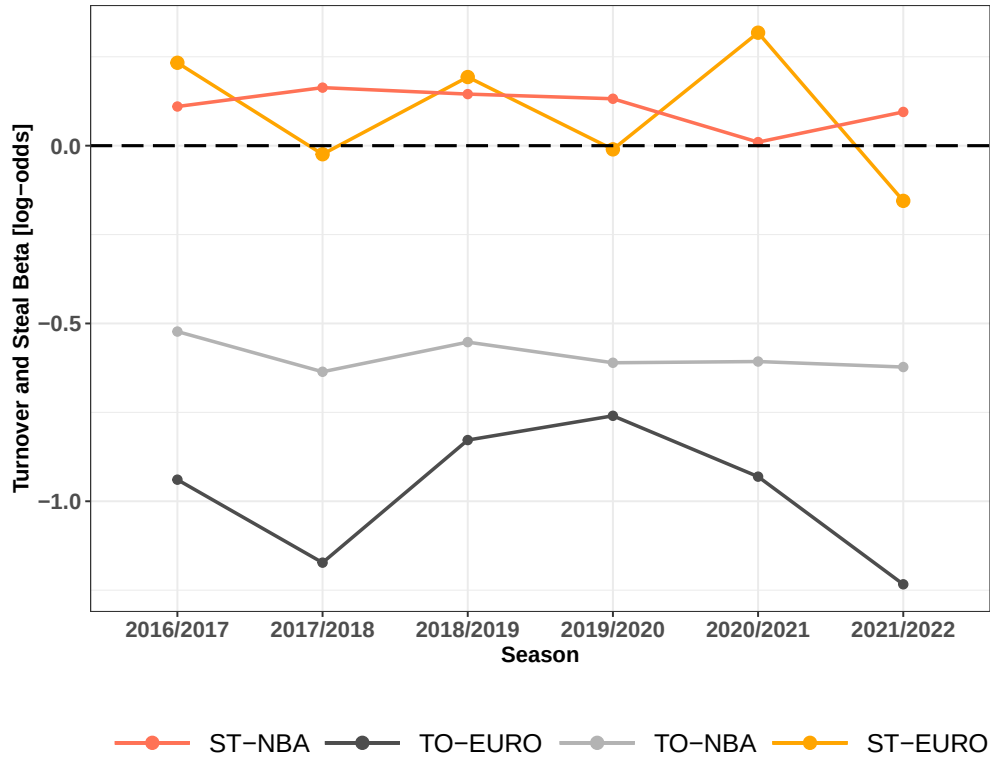


Figure 6.4: Turnover and steal coefficient development between season 2016/2017 and 2021/2022 in Euroleague and NBA

6.4.3 Offensive rating

An analysis of *ORtg* coefficients based on model 1 set of predictors was carried out. Both Euroleague data and the NBA data showed trend in the coefficients of *ORtg* overtime. While in the NBA the trend was not as obvious and rather downwards, the Euroleague data showed a large increase in the importance of offensive efficiency between 2017/2018 and 2021/2022 seasons. On the other hand, the NBA offensive rating's coefficient fluctuated around a value, with signs of decreasing importance in the season 2020/2021 and 2021/2022.

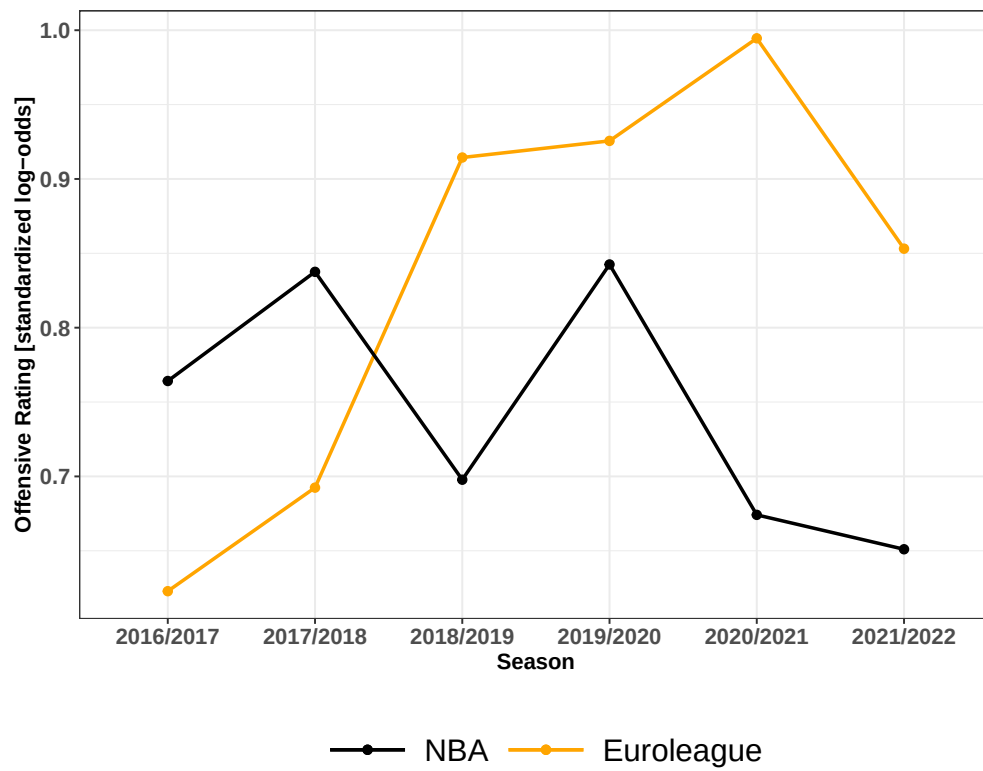


Figure 6.5: $Ortg$ coefficient development between season 2016/2017 and 2021/2022 in Euroleague and NBA

7. Results for Playoff

Similar to regular season, three data sets were created containing games and statistic for every playoff game played between 2016/2017 season and 2021/2022 season. Due to different post season formats in each of the three leagues, a different number of playoff games were played each season. Again, models based on the three sets of predictors were created. The three sets of predictors were the same as in regular season models. Also, the 2019/2020 season was omitted in this data set, because there were no playoff games played that year due to Covid-19 pandemic. In this chapter, results of regular season models will be compared with results from the playoff models. Full model tables can be found in the Appendix A.

7.1 Model 1 - Offensive and Defensive rating

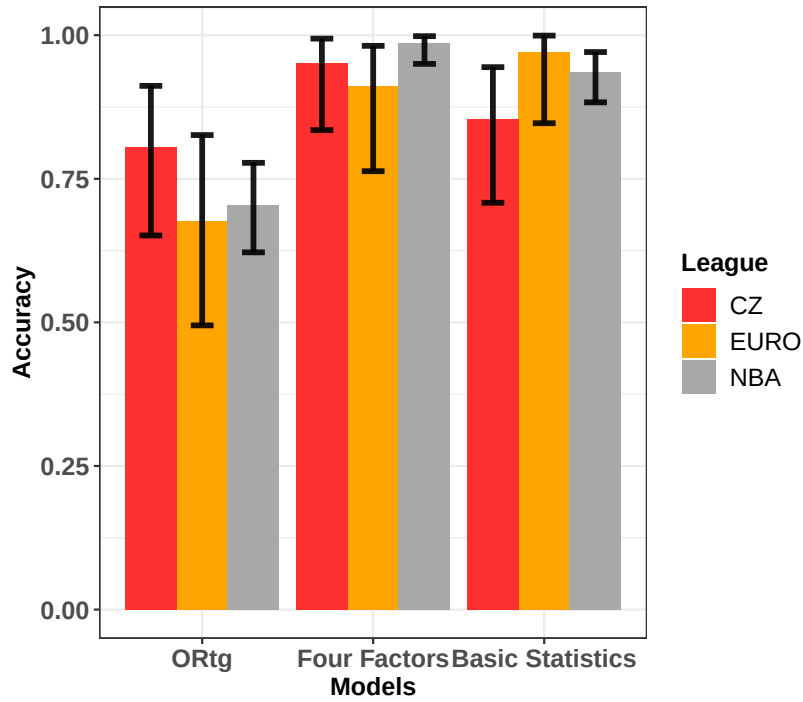
Same as before, the set of predictors containing offensive (defensive) efficiency was used for the first regression model for playoff. The variable was standardized using z scores. Linearity of logit was tested and confirmed by none of the interaction term being significant ($p > 0.1$). Since only one predictor variable was used, no multicollinearity analysis was carried out. An independence between residuals was confirmed as well, and no outliers exceeding Cook's distance of 1 were observed for the NBA and Czech NBL. For Euroleague, one observation exceeded Cook's distance of 1 and was removed from the model's dataset. The accuracy for this model was not again high, with no model exceeding prediction accuracy of 0.8 (Figure 7.1). However, the confidence intervals were wider than for regular season. This was surely caused by the smaller sample size, as leagues with larger sample size (e.g. NBA) had narrower CI.

Table 7.1: Model 1 - Logistic Regression Analysis of Offensive/Defensive rating on Winning a Game in Playoff

	<i>Dependent variable: Win</i>		
	$\hat{\beta}$ Czech NBL (95% CI)	$\hat{\beta}$ Euroleague (95% CI)	$\hat{\beta}$ NBA (95% CI)
ORtg	16.486** (3.328, 29.643)	8.493*** (3.397, 13.589)	9.203*** (6.462, 11.944)
Observations	138	113	498
Log Likelihood	-6.845	-9.128	-41.473
Akaike Inf. Crit.	15.691	20.255	84.946

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$



Note: Error bars = 95% CI

Figure 7.1: Analysis of Accuracy from all models fitted for Playoff

7.1.1 Results

All variables were statistically significant ($p < 0.01$ or $p < 0.05$). The Czech NBL estimate was unusually large with large standard error; therefore large 95% CI. The lower bound of the interval was similar to the Euroleague, but it was the upper bound that was unusually large. This was probably caused by the lower number of observation from the Czech dataset or by the diversity of the Czech NBL playoff series (Table 7.1).

Interestingly, the coefficient for *ORTg* in the NBA was higher than for Euroleague. Comparing this to the regular season (Table 6.2), the importance of offensive (defensive) efficiency in the NBA increased in the playoff, while it decreased in Euroleague playoff. This suggests a trend that offensive style changes in both league in the post season. While in the NBA more emphasis would be on playing efficiently on offence and not allowing the opponent to score efficiently, in the Euroleague the emphasis shifts from efficient shooting somewhere else. Also, the coefficients for NBA and Euroleague are closer together, which supports the hypothesis from (Mandić et al., 2019-10-07) that Euroleague and NBA style of play tend to become similar in the playoff.

To examine other statistics which might play a role in winning basketball games in playoff as well as improve model prediction accuracy, other sets of predictors, like in the regular season were used further.

7.2 Model 2 - Four Factors

The second model utilized Four Factors in predicting the outcome of variable *Win*. All predictors were standardized using z scores. The linearity in logit assumption was not violated, no outliers were detected (Cook's distance < 1), residuals were independently distributed, but multicollinearity analysis showed high values of VIF. For the NBA data, the VIF exceeded 10 for predictors *eFG%*, *TPP* and for Euroleague dataset the VIF exceeded 9 in case of *eFG%*. These variables were not removed from the models, as if they were to be removed, the prediction accuracy would have decreased significantly and the AIC score would have been almost 3 times as large. Also, correlation between the predictors did not exceed an absolute value of 0.3. All these facts considered, the predictors were left in the model, but a possible inflation of coefficients for *eFG%* and *TPP* was kept in mind throughout the interpretation of the models.

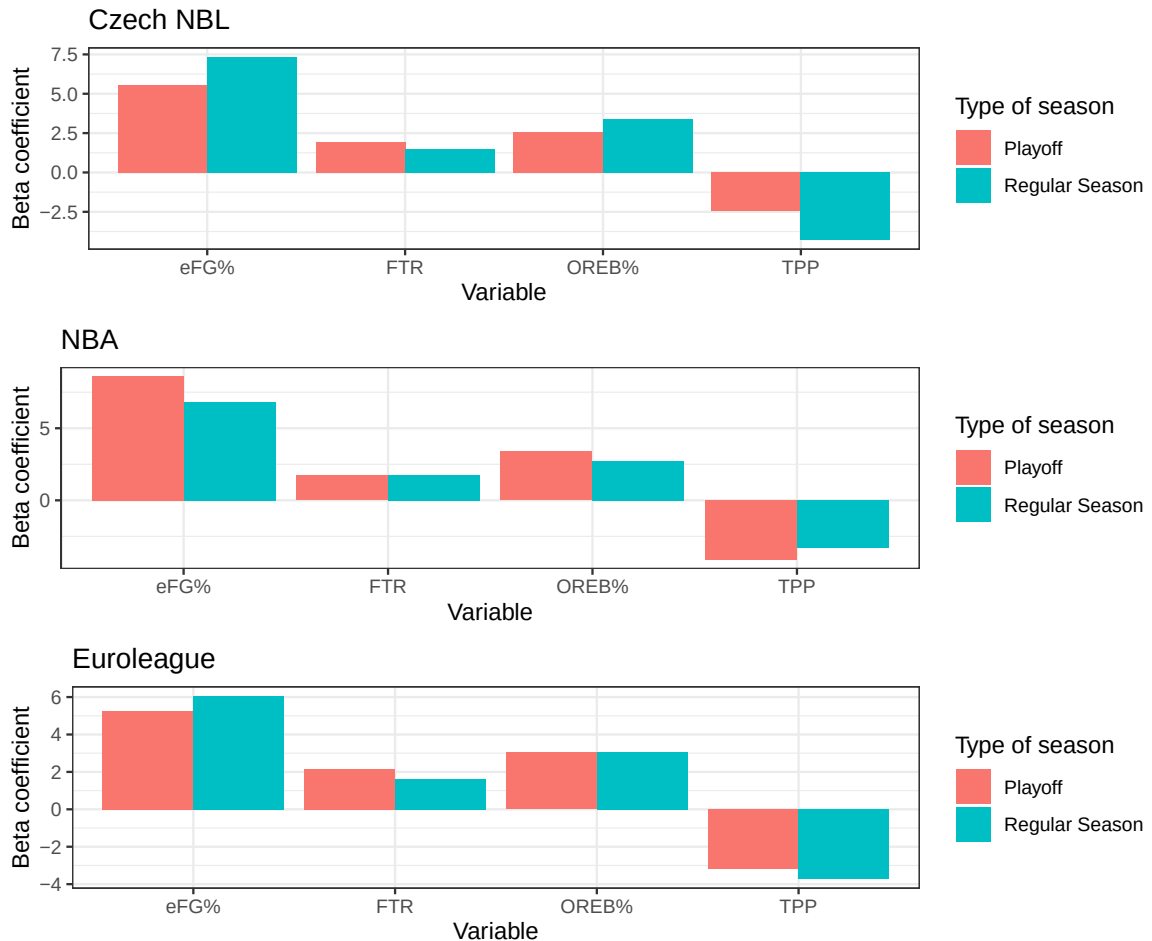


Figure 7.2: Logistic Regression Analysis of Four Factors on Winning a Game in Regular Season and Playoff

7.2.1 Results

Coefficients for playoff and regular season are compared in Figure 7.2. All predictors were found statistically significant in both regular season and playoff. In all three leagues, eFG% was the most influential to winning a game, both in regular season and playoff. However, only in the NBA did the importance of efficient shooting rise. In both Czech NBL and Euroleague the importance of efficient shooting decreased. Besides, keeping in mind, that the coefficient for NBA was inflated ($VIF = 11.2$), it is possible that even in the NBA the importance of efficient shooting decreased in the playoff.

While in the European competitions, the importance of free-throws rose in playoff, it was not the case in the NBA. There, the coefficient of the variable FTR was slightly lower in the playoff. On the other hand, the importance of turnovers and rebounding increased in the NBA post season. Both increased by almost 25% (this can be seen from the table in Appendix A.3). More emphasis on rebounding and handling the ball on offence suggest a different, probably more Europe-like, style of basketball in the NBA playoff (Mandić et al., 2019-10-07).

Turnovers per possessions remained statistically significant and, apart from Czech NBL, the second most important factor from this set of predictors (Appendix A.3). This means that it was still valuable for teams in playoff to not lose control of the ball. Meanwhile, in the Czech NBL, importance of turnovers was overtaken by *OREB%*. This is interesting as it signalizes, that in the Czech playoff it was more important to have offensive rebounds than not turnover the ball. Something similar was observed in the Euroleague, where TPP were still more important than *OREB%*, but the gap narrowed significantly, and offensive rebound were almost as important as turnovers to winning/losing a game in Europe.

7.3 Model 3 - Basic Statistics

Lastly, models fitted with basic statistics as variables were analysed. Variables were not standardized in this case, for better interpretability of the results. For Czech NBL and Euroleague the linearity in logit assumption was not violated. However, for the NBA, the interaction term $ST * \log(ST)$ was statistically significant ($p < 0.01$). This signalizes a non-linear relationship between *ST* and the logit transform of the dependent variable *Win*. Since, previous models with same set of predictors for regular season and models for other leagues in playoff did not seem to violate this assumption and for interpretability and comparison with previous models, this variable will be kept in the model, but will not be interpreted for this particular model. No other assumptions were violated.

Similar to regular season, the prediction accuracy on this set of predictors was high for Euroleague and the NBA, but was notably lower for Czech NBL. Overall, the prediction accuracy was still adequate, with the lowest prediction accuracy being 0.85 (Figure 7.1).

7.3.1 Results - Residual Analysis

Multicollinearity, outliers, and independence of errors assumptions were analysed and none were violated. Nevertheless, in each dataset, at least one observation with high residual appeared. Similar to regular season, these outliers had a common attributes:

- The difference in scores was always 3 points.
- In each game, the home team lost.
- The losing team outperformed the winner in almost every category, except:
 - in Euroleague the winner did better in *3PT%*, *OR*, *AS*,
 - in the NBA the winner had more *AS* and *OR*,
 - in the Czech NBL the winner was better in *3PT%* and *ST*.

This again shows a pattern of offensive rebounds and assists being important in close games, which was apparently not taken into account in this model.

7.3.2 Results - Field Goals

In the previous model, it was shown that in the NBA the importance of efficient shooting rose in the playoff. This model brought more detail on that aspect. Variable *eFG%* consists of two shooting statistics: *2PT%* and *3PT%*. In Figure 7.3 it can be seen that it is mainly 3-point percentage importance that rose in the playoff. The 2-point percentage rose as well, but the overall increase in the coefficient was smaller than for the 3-point shot. The rise in importance of efficient shooting might be due to game slowing down in the playoff. So with slower pace, teams have fewer possessions to score and therefore value high scoring efficiency both from 2 and 3-point range. Table 7.2 compares average number of possessions in regular season and playoff. It shows that NBA teams slow down the most in the post season, on the other side, Czech and Euroleague teams play with the more similar pace to regular season. This could explain the increase in shooting efficiency in the NBA compared to Czech NBL and Euroleague.

7.3.3 Results - Rebounds

As the game slows down, the importance of securing a rebound was increased in playoff in all three observed leagues (Fig. 7.3, Tab. 7.2). Offensive rebounds were statistically significant in all leagues. Defensive rebounding was a statistical significant predictor in all leagues, except Czech NBL (in Euroleague $p < 0.1$) (Table A.4). In Europe-based competitions, offensive rebounding was more important than defensive rebounding, even though, compared to the regular season, the gap between regular season and playoff was smaller. In the NBA, defensive rebounding became more important than offensive rebounding. More emphasis on defence overall was a common theme in the NBA playoff (Mandić et al., 2019-10-07). Again, in Europe, not much difference was observed between regular season and playoff.

Table 7.2: Average number of possessions in Regular Season and Playoff

League	Average number of possessions		Difference
	Playoff	Regular Season	
Czech NBL	76.385	77.237	-0.852
Euroleague	70.373	71.358	-0.986
NBA	99.135	101.447	-2.312

7.3.4 Results - FT%

Free-throw success was insignificant in Czech NBL, however, seemed like a significant contributor to team's success in NBA and Euroleague playoff games. Euroleague playoff games tend to be the closest in final score and Czech playoff games the most lob sided (Tab. 7.3). In closer games, it is the free throws that decide the game, as losing teams are often fouling intentionally. If the game is not close (like in Czech NBL), the importance of free throw percentage is lower than in closer games (like in Euroleague).

Table 7.3: Mean score difference in Playoff

League	Mean	SD
Czech NBL	16.232	12.150
Euroleague	9.763	7.068
NBA	12.594	8.784

7.3.5 Results - Turnovers

Turnovers were again the most important factor in winning a given game. However, comparing regular season and playoff, the importance of turnovers in both Czech NBL and Euroleague decreased. On the contrary, the coefficient rose in the NBA playoff compared to the regular season. This comes with the lowered game pace and more pressure on retaining the ball and committing fewer turnovers, which could have led to opponent scoring. If a team turns the ball over in the NBA playoff, it hurts their odds of winning more than in the regular season. The focus in the NBA shifts from fast-paced not so strategic basketball in regular season to slower tactics-oriented basketball in playoff.

Steals became a statistically significant factor in the Euroleague playoff. They were one of the most important factors, ranking 3rd most important factor behind turnovers and offensive rebounds. However, the 95% CI was relatively wide, so the true effect of steals might be lower, considering their effect in other leagues and regular season.

Both blocks and assists were not statistically significant in any of the models. This might be due to lower number of observations than in regular season. With assists, the reason might be due to a correlation with both $2PT\%$ and $3PT\%$, which is causing assists to not show as

statistically significant. However, from previous researches, which used different method, it is apparent, that passes and assists are contributors to team's success (Ruano et al., 2006-06-01; Ibáñez et al., 2008-11).

Generally, it seemed that while in Euroleague and Czech NBL, the factors important to winning a game did not differ much between regular season and playoff, in the NBA there were some notable differences, indicating NBA teams, in the playoff, play more strategic, slower, maybe even more defensively oriented basketball. The European competitions did not change their style of play in the post-season as much. This might be due to the number of games teams play in different leagues. While in the Euroleague and Czech NBL, teams play about twice a week (around 30 regular season games), NBA teams play 82 regular season games in the similar time span, therefore they play on average 4–5 games a week. This way there is not enough time to prepare, mentally, tactically or physically for every game in the NBA. In Europe, there is more time to prepare for each game and the teams have the luxury to prepare for each game like it is a playoff game.

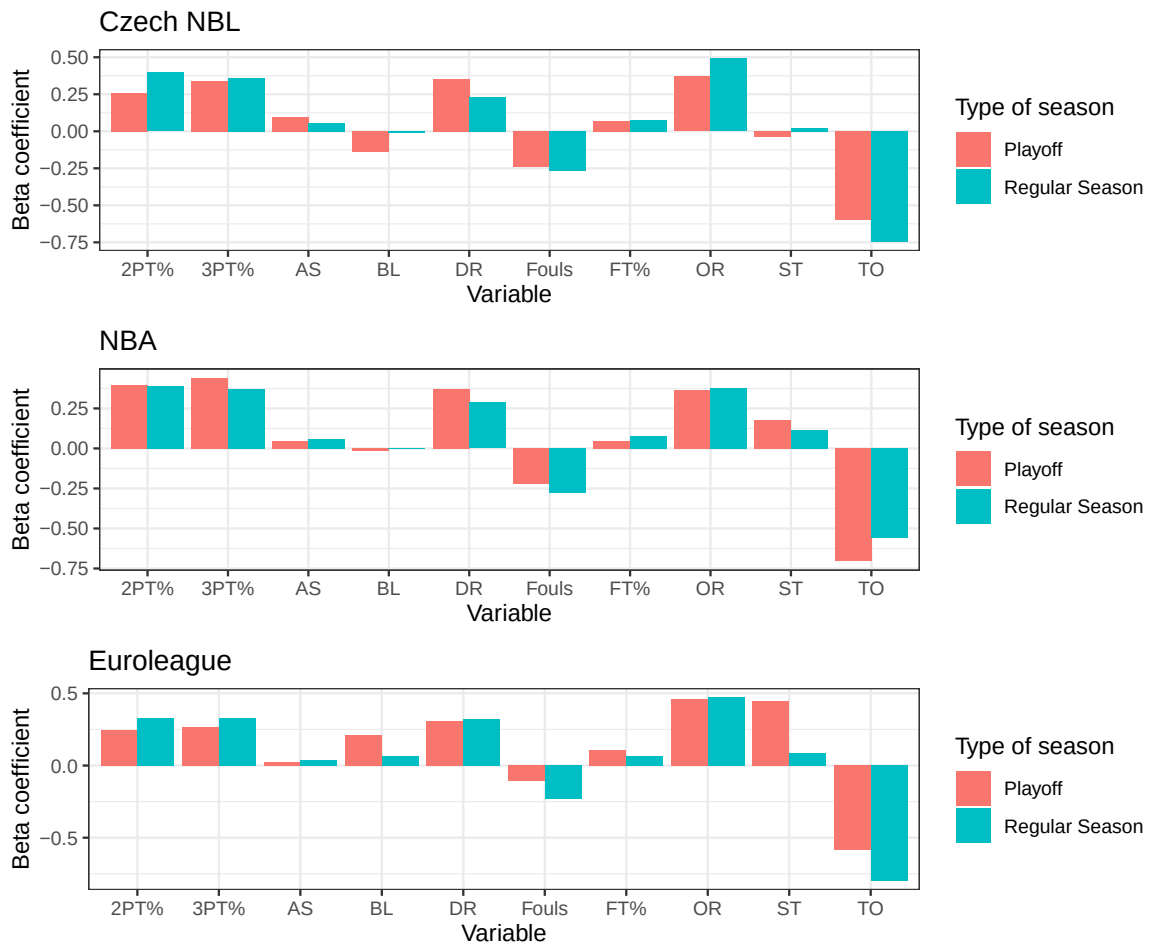


Figure 7.3: Logistic Regression Analysis of Basic Statistics on Winning a Game, Regular Season and Playoff

8. Discussion

The aim of this thesis has been accomplished; the logistic regression models have been fitted and their results interpreted. Nevertheless, this thesis does have limitations, concerning both the methods used and the models themselves.

This thesis used logistic regression models to explain variable win; a variable indicating whether a team won a particular game. This approach takes into consideration only one game success, but not the whole season success. As (Ibáñez et al., 2008-11) points out, even the worst teams in all leagues manage to win some games, and even the best teams lose some games every season. Ultimately, all teams care about is not winning one game, but winning more games than the other teams in the league and winning a championship in the end. Therefore, another approach would be to examine statistic which impact the overall success of a team in a season and not just one game. This way, another types of analysis might had to be used. In (Teramoto and Cross, 2010-01-19) a season long success was measured by team's win percentage and instead of logistic regression, multiple linear regression models were fitted, this was applied to only NBA data. Another approach would be to use linear discriminant analysis, which was applied to NBA data in (Ibáñez et al., 2008-11). There, the dependent variable was if a team did or did not make the playoff that particular season.

One of the main limitations, which was already mentioned in the thesis, is that all created models are not prediction models, as is common for logistic regression. The independent variables in this model are statistics which are known only after the game ends and are difficult to predict themselves. Nonetheless, prediction is not a goal of this thesis. Its goal is only a description of reality and its interpretation.

Lastly, this model used only basic statistics in order to compare different leagues with each other. There are a few limitations connected with using only basic box score statistics. The basic statistics are not able to fully capture what is happening on the basketball court, as there are numerous more actions happening on the court then only a few basic actions captured by the statistics. Actions such as movement of players on offence, good passes, deflected passes on defence or even good communication and team's chemistry might play a smaller or larger role in winning basketball games, but are not captured in the basic game box score. While some of those actions are being counted in the NBA as statistics, no data are available for Euroleague and Czech NBL.

Conclusion

This thesis compared 3 different basketball leagues - Czech NBL, Euroleague and NBA; and attempted to describe the different style of play in each one. It analysed basic and slightly advanced statistics, obtainable or computable from a basic box score of a game.

While some papers and researches have tried to identify the importance of different statistic to winning a basketball game in the NBA and Euroleague, nothing has yet been written regarding the Czech NBL, neither a comparison of the three leagues has been carried out.

The aim of the analysis was identification and description of effect each statistic has on winning or losing the particular game. For this purpose, multiple logistic regression models were created, each based on a different set of predictors. Results from the logistic regression were then compared between the regular season and playoff, as well as compared between the 3 leagues.

At the beginning of this thesis, a set of hypothesis and expectation was established. The subsequent analysis then aimed to confirm or disprove the hypotheses. In doing so, other interesting result came up and were interpreted.

The results of the analysis confirmed some of the hypotheses. In the regular season, all statistics perceived as „good“ had a positive impact on winning a game in all three leagues, on the other side, all statistics perceived as „bad“ had a negative influence on the game outcome. For playoff, however, some statistics such as assists and blocks were not significant, which could be caused by the lower number of observations.

The importance of defensively oriented statistics increased in the playoff. Mainly, it was the defensive rebound importance that increased in the playoff. Also in the Euroleague, the importance of steals increased dramatically compared to the regular season.

The emergence of the 3-point shot and the hypothesised trend, that 3-point shots are becoming more important than 2-point shots, was confirmed, both overall and for the NBA. The value of three point shot has been rising in all three leagues, but mainly in the NBA. There, it is expected that the importance of shooting 3-pointers with high efficiency will overcome the importance of efficient scoring from the 2-point range efficiently in the upcoming seasons. Simultaneously, it was found that the importance of 3-point efficiency is in the NBA significantly higher than in the other two leagues.

With lowered importance of shooting, it was expected that assists would play a key role in Europe-based competitions (Euroleague and Czech NBL). This was not proven right as in all models, assists were statistically insignificant predictor. However, the hypothesis of more team-oriented and tactical basketball in Europe was proved right, as rebounding, turnovers and steal were more important in Euroleague and Czech NBL than they were in the NBA. On the other hand, shooting efficiency was more important in the NBA than in the other

two leagues.

Surprisingly, the most important statistics were turnovers and offensive rebounding in the regular season, with a shift towards defensive rebounding in the playoff. Turnovers were always the most important factor in winning games. Only the efficient field percentage in the Four Factors models was more important. A surprise was the importance of offensive rebounding, which were among the most important factors in winning games in all three leagues.

Apart from providing conclusions to the established hypotheses, this thesis provided other interesting findings not expected prior to the analysis. The most important observation was the difference in the style of play in the NBA and Europe. Generally, in the NBA the key factor to winning a game was efficient shooting, in the Euroleague shooting efficiency was not as important and other statistics (such as rebounding and turnovers) were more important. The Czech NBL was somewhere in the middle, combining the European style of play observed in Euroleague with the NBA style of play, influenced by a high number of foreign players in the Czech NBL (especially from the USA).

A difference between playoff and regular season in different leagues was observed. In all three leagues, the pace in playoff slowed down, and the importance shifted from offensive rebounds towards defensive rebounds. The differences between playoff and regular season were more notable in the NBA, where numerous statistics gained or lost importance in the post season. In Euroleague and Czech NBL, there was some difference between regular season and playoff, but the differences were not as large as in the NBA.

Apart for serving academical purposes, this thesis can also serve to coaches and players participating or preparing to participate in any of the three leagues. This thesis compared the different play style in each league and tried to provide explanation of the differences. Due to different play styles, different statistics were more or less important in each league. This thesis tried to highlight and explain these differences. This could also provide some insight into which skills (projected into statistics) are important in each league, and therefore could help players and coaches prepare for participation or transition from one league to another.

Bibliography

- Advanced stats for everybody* [Hack a Stat], 2023 [online]. [visited on 2023-04-07]. Available from: <https://hackastat.eu/en/home-page-eng/>.
- BASKETBALL, CZ, 0209-08-10. *Jen 17 Američanů! A kdo nejvíc pouštěl žilou? / NBL.BASKETBALL* [CZ BASKETBALL] [online]. 0209-08-10. [visited on 2023-04-07]. Available from: <https://nbl.basketball/jen-17-americanu-a-kdo-nejvic-poustel-zilou/1466>.
- BASKETBALL, CZ, 2021. *ORGANIZACE, SYSTÉM ŘÍZENÍ A SPORTOVNĚ TECHNICKÉ PODMÍNKY PRO ÚČAST V MISTROVSTVÍ ČR V BASKETBALE MUŽŮ*. CZ.BASKETBALL. Available also from: https://cz.basketball/upload/docs/1632995113_STPmanual_WEB_A4.pdf.
- BASKETBALL, CZ, 2022-10-28. *Jak kdo těží z cizinců? A kdo má ty nej? / NBL.BASKETBALL* [online]. 2022-10-28. [visited on 2023-04-07]. Available from: <https://nbl.basketball/jak-kdo-tezi-z-cizincu-a-kdo-ma-ty-nej/14302>.
- BRESLER, Alex, 2023. *nbastatR: R's interface to NBA data*. Available also from: <https://github.com/abresler/nbastatR>. R package version 0.1.152.
- Competition Format 2022-23* [Euroleague Basketball], 2022 [online]. [visited on 2023-04-19]. Available from: <https://www.euroleaguebasketball.net/euroleague/format-el/>.
- CZ.BASKETBALL, 2023 [online]. [visited on 2023-04-07]. Available from: <https://cz.basketball/>.
- DEMARIS, Alfred, 1995. A Tutorial in Logistic Regression. *Journal of Marriage and Family* [online]. Vol. 57, no. 4, pp. 956–968 [visited on 2023-04-07]. ISSN 0022-2445. Available from DOI: 10.2307/353415. Publisher: [Wiley, National Council on Family Relations].
- Differences between NBA and FIBA rules* [basketnews.com], 2022-08-05 [online]. 2022-08-05. [visited on 2023-04-18]. Available from: <https://basketnews.com/news-176033-differences-between-nba-and-fiba-rules.html>.
- DOHMEN, Thomas and SAUERMAN, Jan, 2016-09-01. Referee Bias. *Journal of Economic Surveys*. Vol. 30, pp. 679–695. Available from DOI: 10.1111/joes.12106.
- ENTINE, Oliver A. and SMALL, Dylan S., 2008-04-28. The Role of Rest in the NBA Home-Court Advantage. *Journal of Quantitative Analysis in Sports* [online]. Vol. 4, no. 2 [visited on 2023-04-09]. ISSN 1559-0410. Available from DOI: 10.2202/1559-0410.1106. Publisher: De Gruyter.
- EuroLeague in distance, 2022. *EuroLeague in distance: lucky Italians and biggest travellers from Tel Aviv* [basketnews.com] [online]. [visited on 2023-04-07]. Available from: <https://basketnews.com/news-175514-euroleague-in-distance-lucky-italians-and-biggest-travellers-from-tel-aviv.html>.

- EuroLeague registers all-time attendance, competitiveness records at mid-season / EuroLeague* [Euroleague Basketball], 2023-01-18 [online]. 2023-01-18. [visited on 2023-04-07]. Available from: <https://www.euroleaguebasketball.net/euroleague/news/euroleague-registers-all-time-attendance-competitiveness-records-at-mid-season/>.
- FREITAS, Lucas, 2021-06-01. Shot distribution in the NBA: did we see when 3-point shots became popular? *German Journal of Exercise and Sport Research* [online]. Vol. 51, no. 2, pp. 237–240 [visited on 2023-04-07]. ISSN 2509-3150. Available from DOI: 10.1007/s12662-020-00690-7.
- GOUGH, Christina, 2022-06-23. *NBA average attendance (regular season) 2022* [Statista] [online]. 2022-06-23. [visited on 2023-04-07]. Available from: <https://www.statista.com/statistics/193632/average-regular-season-home-attendance-per-team-in-the-nba-since-2006/>.
- GRÖER, Dr Chris, 2017-02-15. *KPMG BrandVoice: Upping The Scheduling Game For Professional Sports* [Forbes] [online]. 2017-02-15. [visited on 2023-04-13]. Available from: <https://www.forbes.com/sites/kpmg/2017/02/15/upping-the-scheduling-game-for-professional-sports/>. Section: Business.
- How the NBA Schedule is Made* [NBASTuffer], 2017-05-08 [online]. 2017-05-08. [visited on 2023-04-18]. Available from: <https://www.nbastuffer.com/analytics101/how-the-nba-schedule-is-made/>.
- IBÁÑEZ, Sergio J., SAMPAIO, Jaime, FEU, Sebastian, LORENZO, Alberto, GÓMEZ, Miguel A. and ORTEGA, Enrique, 2008-11. Basketball game-related statistics that discriminate between teams' season-long success. *European Journal of Sport Science* [online]. Vol. 8, no. 6, pp. 369–372 [visited on 2023-04-09]. ISSN 1746-1391, ISSN 1536-7290. Available from DOI: 10.1080/17461390802261470.
- KUBATKO, Justin, OLIVER, Dean, PELTON, Kevin and ROSENBAUM, Dan T, 2007-01-09. A Starting Point for Analyzing Basketball Statistics. *Journal of Quantitative Analysis in Sports* [online]. Vol. 3, no. 3 [visited on 2023-04-09]. ISSN 1559-0410. Available from DOI: 10.2202/1559-0410.1070.
- LORENZO, Jorge, LORENZO, Alberto, CONTE, Daniele and GIMÉNEZ, Mario, 2019. Long-Term Analysis of Elite Basketball Players' Game-Related Statistics Throughout Their Careers. *Frontiers in Psychology* [online]. Vol. 10 [visited on 2023-04-07]. ISSN 1664-1078. Available from: <https://www.frontiersin.org/articles/10.3389/fpsyg.2019.00421>.
- MANDIĆ, Radivoj, JAKOVLJEVIĆ, Saša, ERČULJ, Frane and ŠTRUMBELJ, Erik, 2019-10-07. Trends in NBA and Euroleague basketball: Analysis and comparison of statistical data from 2000 to 2017. *PLoS ONE* [online]. Vol. 14, no. 10, e0223524 [visited on 2023-04-07]. ISSN 1932-6203. Available from DOI: 10.1371/journal.pone.0223524.
- MCMAHAN, Ian, 2018-10-26. How sleep and jet-lag influences success in the travel-crazy NBA. *The Guardian* [online] [visited on 2023-04-13]. ISSN 0261-3077. Available from: <https://www.theguardian.com/sport/2018/oct/26/sleep-nba-effects-basketball>.

- MINAROVIČOVÁ, Kristýna, 2014. *Zahraniční hráči v nejvyšších českých basketbalových soutěžích mužů a žen* Bakalářská práce. Brno. Available also from: https://is.muni.cz/th/q8x3g/Minarovicova_Kristyna_Bakalarska_prace_tisk.pdf. Bakalářská práce. Masarykova Univerzita.
- NBA playoffs*, 2023-04-17. In: *Wikipedia* [online] [visited on 2023-04-18]. Available from: https://en.wikipedia.org/w/index.php?title=NBA_playoffs&oldid=1150329715. Page Version ID: 1150329715.
- NEVILL, Alan and HOLDER, R, 1999-11-01. Home advantage in sport: an overview of studies on the advantage of playing at home. *Sports medicine (Auckland, N.Z.)* Vol. 28, pp. 221–36.
- R CORE TEAM, 2021. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. Available also from: <https://www.R-project.org/>.
- REDICK, JJ and ALTER, Tommy, 2023-03-30. *The Old Man & the Three with JJ Redick and Tommy Alter* [online]. [visited on 2023-04-13]. No. 152. Available from: <https://open.spotify.com/episode/5UNHgFmPkQeCz6C8tIBWNP>.
- RUANO, Miguel, EVANGELOS, Tsamourtzis and LORENZO CALVO, Alberto, 2006-06-01. Defensive systems in basketball ball possessions. *International Journal of Performance Analysis in Sport*. Vol. 6, pp. 98–107. Available from DOI: 10.1080/24748668.2006.11868358.
- SCHUHMANN, John, 2021-10-14. *NBA's 3-point revolution: How 1 shot is changing the game* [NBA.com] [online]. 2021-10-14. [visited on 2023-04-13]. Available from: <https://www.nba.com/news/3-point-era-nba-75>.
- STOLTZFUS, Jill C., 2011-10. Logistic Regression: A Brief Primer: LOGISTIC REGRESSION: A BRIEF PRIMER. *Academic Emergency Medicine* [online]. Vol. 18, no. 10, pp. 1099–1104 [visited on 2023-04-21]. ISSN 10696563. Available from DOI: 10.1111/j.1553-2712.2011.01185.x.
- TABACHNICK, Barbara G. and FIDELL, Linda S., 2013. *Using multivariate statistics*. 6. ed., international ed. Boston Munich: Pearson. Always learning. ISBN 978-0-205-89081-1 978-1-292-02131-7.
- TERAMOTO, Masaru and CROSS, Chad L., 2010-01-19. Relative Importance of Performance Factors in Winning NBA Games in Regular Season versus Playoffs. *Journal of Quantitative Analysis in Sports* [online]. Vol. 6, no. 3 [visited on 2023-04-09]. ISSN 1559-0410. Available from DOI: 10.2202/1559-0410.1260.
- WEBB, Michael, ARRINGTON, Cole and KAZANOWSKI, Brian, 2012. *NBA vs Euroleague Player Heights* [online]. [visited on 2023-04-07]. Available from: <https://www.statcrunch.com/reports/view?reportid=28650&tab=preview>.
- WOOLDRIDGE, Jeffrey M., 2013. *Introductory econometrics: a modern approach*. 5th ed. Mason, OH: South-Western Cengage Learning. ISBN 978-1-111-53104-1. OCLC: ocn827938223.

ZLÁMEL, Filip, 2013. *Logistická regrese v R*. Brno. Available also from: https://is.muni.cz/th/jkx3p/bakalarska_prace_Filip_Zlamal.pdf. Bakalářská práce. Přírodovědecká fakulta, Masarykova univerzita Ústav matematiky a statistiky.

Appendices

A. Appendix A

A.1 Transformation of coefficients for Model 3 in Regular season

Table A.1: Model 3 transformed - Logistic Regression Analysis of Basic Statistics on Winning a Game in Regular Season

<i>Dependent variable: Win</i>			
	Odds Ratio		
	Czech NBL	Euroleague	NBA
OR	1.636***	1.603***	1.456***
DR	1.259***	1.380***	1.333***
BL	0.990	1.064	1.005
AS	1.059	1.036	1.059***
ST	1.022	1.090	1.123***
TO	0.474***	0.450***	0.571***
Foul	0.765***	0.791***	0.759***
2PT%	1.489***	1.389***	1.474***
3PT%	1.436***	1.391***	1.453***
FT%	1.080***	1.068***	1.081***
Observations	789	1,498	7,059
Log Likelihood	−83.665	−205.576	−1,031.040
Akaike Inf. Crit.	187.330	431.153	2,082.080

Note:

*p<0.1; **p<0.05; ***p<0.01

A.2 Table of Results for Logistic Regression Analysis of Model 1, Regular Season and Playoff

Table A.2: Logistic Regression Analysis of ORtg and DRtg on Winning a Game - comparison of Regular Season and Playoff

	<i>Dependent variable: Win</i>					
	CZ-PL	EURO-PL	NBA-PL	CZ-RS	EURO-RS	NBA-RS
	$\hat{\beta}$ (95% CI)	$\hat{\beta}$ (95% CI)	$\hat{\beta}$ (95% CI)	$\hat{\beta}$ (95% CI)	$\hat{\beta}$ (95% CI)	$\hat{\beta}$ (95% CI)
ORtg	16.486** (3.328, 29.643)	8.493*** (3.397, 13.589)	9.203*** (6.462, 11.944)	9.098*** (6.889, 11.307)	10.781*** (8.883, 12.679)	8.282*** (7.695, 8.870)
Observations	138	113	497	789	1,498	7,059
Log Likelihood	-6.845	-9.128	-41.473	-58.981	-105.204	-698.895
Akaike Inf. Crit.	15.691	20.255	84.946	119.962	212.407	1,399.790

Note:

*p<0.1; **p<0.05; ***p<0.01

$\hat{\beta}$ = standardized coefficients, CI = confidence intervals

CZ = Czech NBL, EURO = Euroleague, PL = Playoff, RS = Regular Season

A.3 Table of Results for Logistic Regression Analysis of Model 2, Regular Season and Playoff

Table A.3: Model 2 - Logistic Regression Analysis of Four Factors on Winning a Game, Regular Season and Playoff

	<i>Dependent variable: Win</i>					
	CZ-P $\hat{\beta}$ (95% CI)	CZ-RS $\hat{\beta}$ (95% CI)	NBA-P $\hat{\beta}$ (95% CI)	NBA-RS $\hat{\beta}$ (95% CI)	EURO-P $\hat{\beta}$ (95% CI)	EURO-RS $\hat{\beta}$ (95% CI)
eFG	5.535*** (2.765, 8.305)	7.347*** (5.499, 9.195)	8.604*** (6.050, 11.157)	6.819*** (6.347, 7.291)	5.239*** (2.334, 8.145)	6.068*** (5.181, 6.955)
TPP	-2.447*** (-3.951, -0.944)	-4.304*** (-5.444, -3.164)	-4.119*** (-5.414, -2.824)	-3.275*** (-3.520, -3.031)	-3.203*** (-5.098, -1.307)	-3.734*** (-4.334, -3.134)
OREB	2.583*** (1.204, 3.962)	3.405*** (2.459, 4.350)	3.386*** (2.307, 4.465)	2.749*** (2.540, 2.958)	3.042*** (1.033, 5.052)	3.056*** (2.560, 3.551)
FTR	1.935*** (0.720, 3.149)	1.506*** (1.006, 2.005)	1.726*** (1.056, 2.396)	1.775*** (1.625, 1.925)	2.138*** (0.670, 3.605)	1.623*** (1.306, 1.940)
Observations	138	789	497	7,059	113	1,498
Log Likelihood	-14.508	-57.754	-37.587	-749.386	-15.324	-157.818
Akaike Inf. Crit.	37.015	123.508	83.174	1,506.771	38.648	323.635

Note:

$\hat{\beta}$ = standardized coefficients, CI = confidence intervals

CZ = Czech NBL, EURO = Euroleague, PL = Playoff, RS = Regular Season

*p<0.1; **p<0.05; ***p<0.01

A.4 Table of Results for Logistic Regression Analysis of Model 3, Regular Season and Playoff

Table A.4: Logistic Regression Analysis of Basic Statistics on Winning a Game - comparison of Regular Season and Playoff

	<i>Dependent variable: Win</i>					
	$\hat{\beta}$ CZ-P (95% CI)	$\hat{\beta}$ CZ-RS (95% CI)	$\hat{\beta}$ NBA-P (95% CI)	$\hat{\beta}$ NBA-RS (95% CI)	$\hat{\beta}$ EURO-P (95% CI)	$\hat{\beta}$ EURO-RS (95% CI)
2PT%	0.260** (0.034, 0.485)	0.398*** (0.289, 0.506)	0.394*** (0.268, 0.521)	0.388*** (0.356, 0.419)	0.247*** (0.092, 0.402)	0.328*** (0.268, 0.389)
3PT%	0.339*** (0.128, 0.551)	0.362*** (0.271, 0.453)	0.441*** (0.315, 0.566)	0.374*** (0.348, 0.399)	0.265*** (0.112, 0.418)	0.330*** (0.278, 0.381)
FT%	0.070 (-0.028, 0.168)	0.077*** (0.045, 0.109)	0.047** (0.010, 0.085)	0.078*** (0.068, 0.087)	0.104** (0.025, 0.184)	0.065*** (0.046, 0.085)
AS	0.098 (-0.107, 0.303)	0.057 (-0.024, 0.138)	0.049 (-0.044, 0.143)	0.057*** (0.036, 0.079)	0.023 (-0.191, 0.237)	0.035 (-0.018, 0.089)
ST	-0.037 (-0.442, 0.367)	0.022 (-0.121, 0.165)	0.178 (-0.047, 0.402)	0.116*** (0.072, 0.160)	0.448* (-0.016, 0.912)	0.086 (-0.018, 0.189)
DR	0.353 (-0.150, 0.855)	0.231*** (0.093, 0.368)	0.373*** (0.210, 0.535)	0.287*** (0.251, 0.323)	0.307* (-0.050, 0.663)	0.322*** (0.213, 0.431)
OR	0.369** (0.075, 0.663)	0.492*** (0.342, 0.643)	0.367*** (0.217, 0.516)	0.376*** (0.340, 0.412)	0.460*** (0.121, 0.800)	0.472*** (0.377, 0.567)
Fouls	-0.237* (-0.496, 0.022)	-0.267*** (-0.370, -0.165)	-0.222*** (-0.336, -0.107)	-0.276*** (-0.303, -0.248)	-0.104 (-0.349, 0.140)	-0.234*** (-0.300, -0.168)
BL	-0.135 (-0.623, 0.352)	-0.010 (-0.164, 0.145)	-0.014 (-0.174, 0.146)	0.005 (-0.030, 0.039)	0.214 (-0.196, 0.624)	0.062 (-0.044, 0.168)
TO	-0.596** (-1.067, -0.125)	-0.746*** (-0.954, -0.539)	-0.705*** (-0.960, -0.450)	-0.561*** (-0.612, -0.510)	-0.585*** (-0.983, -0.186)	-0.798*** (-0.944, -0.653)
Observations	138	789	497	7,059	114	1,498
Log Likelihood	-15.407	-83.665	-61.045	-1,031.040	-19.235	-205.576
Akaike Inf. Crit.	50.814	187.330	142.090	2,082.080	58.470	431.153

Note:

*p<0.1; **p<0.05; ***p<0.01

$\hat{\beta}$ = coefficients, CI = confidence intervals

CZ = Czech NBL, EURO = Euroleague, PL = Playoff, RS = Regular Season

A.5 Proof for variable selection

For simplicity, only 1 statistic from both teams is used, therefore 2 variables are in the model, but are in form of interaction:

$$\begin{aligned}
 \pi[Win_t] &= \frac{1}{1 - e^{-\beta(\theta_t - \theta_o)}} = 1 - \pi[Win_o] \\
 &= 1 - \frac{1}{1 - e^{-\beta(\theta_o - \theta_t)}} = 1 - \frac{1}{1 - e^{\beta(\theta_t - \theta_o)}} \\
 &= \frac{1 - e^{\beta(\theta_t - \theta_o)} - 1}{1 - e^{\beta(\theta_t - \theta_o)}} = -\frac{e^{\beta(\theta_t - \theta_o)}}{1 - e^{\beta(\theta_t - \theta_o)}} \\
 &= -\frac{e^{\beta(\theta_t - \theta_o)}}{1 - e^{\beta(\theta_t - \theta_o)}} \frac{e^{-\beta(\theta_t - \theta_o)}}{e^{-\beta(\theta_t - \theta_o)}} \\
 &= -\frac{1}{e^{-\beta(\theta_t - \theta_o)} - 1} = \frac{1}{1 - e^{-\beta(\theta_t - \theta_o)}} \\
 &= \pi[Win_t],
 \end{aligned}$$

where $\pi[Win_t]$ is a Win of a team t , $\pi[Win_o]$ is a win of their opponent, θ_t is some statistic of a team t and θ_o is the same statistic for their opponent.

The same is not true for the model without the interaction term:

$$\pi[Win_t] = \frac{1}{1 - e^{-\beta_1\theta_t + \beta_2\theta_o}} \neq 1 - \pi[Win_o]$$